



284428991 Electromagnetics Drill Solution Hayt8e Chapter 1to5

Engineering Electromagnetics by William Hyatt-8th Edition (Ghulam Ishaq Khan Institute
of Engineering Sciences and Technology)

ELECTROMAGNETICS HAYT 8 EDITION

DRILL SOLUTION FROM CHAPTERS 1 TO 5

CHAPTER 1

D1.1. Given points $M(-1, 2, 1)$, $N(3, -3, 0)$, and $P(-2, -3, -4)$, find: (a) $\mathbf{R}_{MN}$; (b) $\mathbf{R}_{MN} + \mathbf{R}_{MP}$; (c) $|\mathbf{r}_M|$; (d) $\mathbf{a}_{MP}$; (e) $|2\mathbf{r}_P - 3\mathbf{r}_N|$.

$$(a) \mathbf{R}_{MN} = (3 - (-1))\mathbf{a}_x + (-3 - 2)\mathbf{a}_y + (0 - 1)\mathbf{a}_z = 4\mathbf{a}_x - 5\mathbf{a}_y - \mathbf{a}_z$$

$$(b) \mathbf{R}_{MN} + \mathbf{R}_{MP} = \mathbf{R}_{MN} + (-2 - (-1))\mathbf{a}_x + (-3 - 2)\mathbf{a}_y + (-4 - 1)\mathbf{a}_z \\ = \mathbf{R}_{MN} - \mathbf{a}_x - 5\mathbf{a}_y - 5\mathbf{a}_z = 4\mathbf{a}_x - 5\mathbf{a}_y - \mathbf{a}_z - \mathbf{a}_x - 5\mathbf{a}_y - 5\mathbf{a}_z \\ = 3\mathbf{a}_x - 10\mathbf{a}_y - 6\mathbf{a}_z$$

$$(c) |\mathbf{r}_M| = \sqrt{(-1)^2 + 2^2 + 1^2} = \sqrt{6} = 2.45$$

$$(d) \mathbf{a}_{MP} = \frac{-\mathbf{a}_x - 5\mathbf{a}_y - 5\mathbf{a}_z}{\sqrt{(-1)^2 + (-5)^2 + (-5)^2}} = \frac{-\mathbf{a}_x - 5\mathbf{a}_y - 5\mathbf{a}_z}{\sqrt{51}} = -0.14\mathbf{a}_x - 0.7\mathbf{a}_y - 0.7\mathbf{a}_z$$

$$(e) 2\mathbf{r}_P - 3\mathbf{r}_N = 2(-2\mathbf{a}_x - 3\mathbf{a}_y - 4\mathbf{a}_z) - 3(3\mathbf{a}_x - 3\mathbf{a}_y) = -4\mathbf{a}_x - 6\mathbf{a}_y - 8\mathbf{a}_z - 9\mathbf{a}_x + 9\mathbf{a}_y \\ = -13\mathbf{a}_x + 3\mathbf{a}_y - 8\mathbf{a}_z$$

$$|2\mathbf{r}_P - 3\mathbf{r}_N| = \sqrt{(-13)^2 + 3^2 + (-8)^2} = 11\sqrt{2} = 15.56$$

D1.2. A vector field $\mathbf{S}$ is expressed in rectangular coordinates as $\mathbf{S} = \{125/[(x-1)^2 + (y-2)^2 + (z+1)^2]\} \{(x-1)\mathbf{a}_x + (y-2)\mathbf{a}_y + (z+1)\mathbf{a}_z\}$. (a) Evaluate $\mathbf{S}$ at $P(2, 4, 3)$. (b) Determine a unit vector that gives the direction of $\mathbf{S}$ at P . (c) Specify the surface $f(x, y, z)$ on which $|\mathbf{S}| = 1$.

$$(a) \mathbf{S} = \frac{125}{(2-1)^2 + (4-2)^2 + (3+1)^2} \{(2-1)\mathbf{a}_x + (4-2)\mathbf{a}_y + (3+1)\mathbf{a}_z\} \\ = \frac{125}{1^2 + 2^2 + 4^2} (\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z) = \frac{125}{21} (\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z) \\ = 5.95\mathbf{a}_x + 11.9\mathbf{a}_y + 23.8\mathbf{a}_z$$

$$(b) \mathbf{a}_S = \frac{5.95\mathbf{a}_x + 11.9\mathbf{a}_y + 23.8\mathbf{a}_z}{\sqrt{5.95^2 + 11.9^2 + 23.8^2}} = \frac{5.95\mathbf{a}_x + 11.9\mathbf{a}_y + 23.8\mathbf{a}_z}{27.27} \\ = 0.218\mathbf{a}_x + 0.436\mathbf{a}_y + 0.873\mathbf{a}_z$$

$$(c) 1 = |\mathbf{S}| =$$

$$\sqrt{\left(\frac{125(x-1)}{(x-1)^2 + (y-2)^2 + (z+1)^2}\right)^2 + \left(\frac{125(y-2)}{(x-1)^2 + (y-2)^2 + (z+1)^2}\right)^2 + \left(\frac{125(z+1)}{(x-1)^2 + (y-2)^2 + (z+1)^2}\right)^2} \\ 1 = \sqrt{\frac{125^2[(x-1)^2 + (y-2)^2 + (z+1)^2]}{[(x-1)^2 + (y-2)^2 + (z+1)^2]^2}} = \frac{125\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}}{(x-1)^2 + (y-2)^2 + (z+1)^2}$$

Transposing,

$$\frac{(x-1)^2 + (y-2)^2 + (z+1)^2}{\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}} = 125$$

Conjugating,

$$\frac{(x-1)^2 + (y-2)^2 + (z+1)^2}{\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}} \cdot \frac{\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}}{\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}} = 125 \\ \sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2} = 125$$

D1.3. The three vertices of a triangle are located at $A(6, -1, 2)$, $B(-2, 3, -4)$, and $C(-3, 1, 5)$. Find: (a) $\mathbf{R}_{AB}$; (b) $\mathbf{R}_{AC}$; (c) the angle θ_{BAC} at vertex A ; (d) the (vector) projection of $\mathbf{R}_{AB}$ on $\mathbf{R}_{AC}$.

$$(a) \mathbf{R}_{AB} = (-2 - 6)\mathbf{a}_x + (3 - (-1))\mathbf{a}_y + (-4 - 2)\mathbf{a}_z \\ = -8\mathbf{a}_x + 4\mathbf{a}_y - 6\mathbf{a}_z$$

$$(b) \mathbf{R}_{AC} = (-3 - 6)\mathbf{a}_x + (1 - (-1))\mathbf{a}_y + (5 - 2)\mathbf{a}_z \\ = -9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

(c) Locating angle θ_{BAC} ,

$$\mathbf{R}_{AB} \cdot \mathbf{R}_{AC} = (-8)(-9) + 4(2) + (-6)(3) = 72 + 8 - 18 = 62$$

Using dot product,

$$\mathbf{R}_{AB} \cdot \mathbf{R}_{AC} = |\mathbf{R}_{AB}||\mathbf{R}_{AC}| \cos \theta_{BAC}$$

$$\theta_{BAC} = \cos^{-1} \frac{\mathbf{R}_{AB} \cdot \mathbf{R}_{AC}}{|\mathbf{R}_{AB}| |\mathbf{R}_{AC}|} = \cos^{-1} \frac{62}{\left(\sqrt{(-8)^2 + 4^2 + (-6)^2}\right) \left(\sqrt{(-9)^2 + 2^2 + 3^2}\right)}$$

$$= 53.6^\circ$$

(d) Analyzing the projection,

$$|\mathbf{R}_{AB(L)}| = \mathbf{R}_{AB} \cdot \mathbf{a}_{AC} = \mathbf{R}_{AB} \cdot \frac{\mathbf{R}_{AC}}{|\mathbf{R}_{AC}|} = \frac{62}{\sqrt{(-9)^2 + 2^2 + 3^2}} = 6.395$$

$$\mathbf{R}_{AB(L)} = |\mathbf{R}_{AB(L)}| \mathbf{a}_{AC} = |\mathbf{R}_{AB(L)}| \left(\frac{\mathbf{R}_{AC}}{|\mathbf{R}_{AC}|} \right) = (6.395) \frac{-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z}{\sqrt{(-9)^2 + 2^2 + 3^2}}$$

$$= -5.94\mathbf{a}_x + 1.319\mathbf{a}_y + 1.979\mathbf{a}_z$$

D1.4. The three vertices of a triangle are located at $A(6, -1, 2)$, $B(-2, 3, -4)$, and $C(-3, 1, 5)$. Find: (a) $\mathbf{R}_{AB} \times \mathbf{R}_{AC}$; (b) the area of the triangle; (c) a unit vector perpendicular to the plane in which the triangle is located.

$$(a) \mathbf{R}_{AB} \times \mathbf{R}_{AC} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ -8 & 4 & -6 \\ -9 & 2 & 3 \end{vmatrix} = [4(3) - (-6)(2)]\mathbf{a}_x + [(-6)(-9) - (-8)(3)]\mathbf{a}_y + [(-8)(2) - 4(-9)]\mathbf{a}_z$$

$$= 24\mathbf{a}_x + 78\mathbf{a}_y + 20\mathbf{a}_z$$

$$(b) A_\Delta = \frac{1}{2} \text{base} \times \text{height} = \frac{1}{2} |\mathbf{R}_{AB} \times \mathbf{R}_{BC}|$$

Finding the direction of vector $\mathbf{R}_{BC}$

$$\mathbf{R}_{BC} = (-3 - (-2))\mathbf{a}_x + (1 - 3)\mathbf{a}_y + (5 - (-4))\mathbf{a}_z = -\mathbf{a}_x - 2\mathbf{a}_y + 9\mathbf{a}_z$$

$$\mathbf{R}_{AB} \times \mathbf{R}_{BC} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ -8 & 4 & -6 \\ -1 & -2 & 9 \end{vmatrix} = 24\mathbf{a}_x + 78\mathbf{a}_y + 20\mathbf{a}_z$$

$$|\mathbf{R}_{AB} \times \mathbf{R}_{BC}| = \sqrt{24^2 + 78^2 + 20^2} = 84$$

$$A_\Delta = (1/2)(84) = 42$$

$$(c) \mathbf{a}_{\mathbf{R}_{AB} \times \mathbf{R}_{AC}} = \frac{24\mathbf{a}_x + 78\mathbf{a}_y + 20\mathbf{a}_z}{84} = 0.286\mathbf{a}_x + 0.928\mathbf{a}_y + 0.238\mathbf{a}_z$$

D1.5. (a) Give the rectangular coordinates of the point $C(\rho = 4.4, \phi = -115^\circ, z = 2)$. (b) Give the cylindrical coordinates of the point $D(x = -3.1, y = 2.6, z = -3)$. (c) Specify the distance from C to D .

(a) Converting cylindrical to rectangular coordinates

Table 1 Cylindrical to rectangular coordinate systems
$x = \rho \cos \phi$
$y = \rho \sin \phi$
$z = z$

$$x = 4.4 \cos -115^\circ = -1.86$$

$$y = 4.4 \sin -115^\circ = -3.99$$

$$z = 2$$

$$C(x = -1.86, y = -3.99, z = 2)$$

(b) Converting rectangular to cylindrical coordinates

Table 2 Rectangular to cylindrical coordinate systems
$\rho = \sqrt{x^2 + y^2}$
$\phi = \tan^{-1} y/x$
$z = z$

$$\rho = \sqrt{(-3.1)^2 + 2.6^2} = 4.05$$

(ϕ is added with 180° which it lies on the second quadrant.)

$$\phi = \tan^{-1} y/x + 180^\circ = \tan^{-1} (2.6/-3.1) + 180^\circ = 140^\circ$$

$$z = -3$$

$$D(\rho = 4.05, \phi = 140^\circ, z = -3)$$

$$(c) \mathbf{CD} = (-3.1 - (-1.86))\mathbf{a}_x + (2.6 - (-3.99))\mathbf{a}_y + (-3 - 2)\mathbf{a}_z$$

$$= -1.24\mathbf{a}_x + 6.59\mathbf{a}_y - 5\mathbf{a}_z$$

$$|\mathbf{CD}| = \sqrt{(-1.24)^2 + 6.59^2 + (-5)^2} = 8.36$$

D1.6. Transform to cylindrical coordinates: (a) $\mathbf{F} = 10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z$ at point $P(10, -8, 6)$; (b) $\mathbf{G} = (2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y$ at point $Q(\rho, \phi, z)$. (c) Give the rectangular components of the vector $\mathbf{H} = 20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z$ at $P(x = 5, y = 2, z = -1)$.

(a) Taking note that the dot product of the unit vectors in cylindrical to rectangular coordinate systems and vice versa

Table 3 Dot products of the unit vectors in cylindrical and rectangular coordinate systems			
	$\mathbf{a}_\rho$	$\mathbf{a}_\phi$	$\mathbf{a}_z$
$\mathbf{a}_x \cdot$	$\cos \phi$	$-\sin \phi$	0
$\mathbf{a}_y \cdot$	$\sin \phi$	$\cos \phi$	0
$\mathbf{a}_z \cdot$	0	0	1

$$\begin{aligned}\mathbf{F} &= (10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z) \cdot \mathbf{a}_\rho + (10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z) \cdot \mathbf{a}_\phi + (10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z) \cdot \mathbf{a}_z \\ &= (10 \cos \phi - 8 \sin \phi + 0)\mathbf{a}_\rho + [(10(-\sin \phi) + (-8)(\cos \phi) + 0)]\mathbf{a}_\phi + (0 + 0 + 6)\mathbf{a}_z\end{aligned}$$

$$\begin{aligned}\text{Since } \phi &= (\tan^{-1}(-8/10) + 180^\circ) = 141.43^\circ \\ &= (10 \cos 141.43^\circ - 8 \sin 141.43^\circ)\mathbf{a}_\rho + [(10(-\sin 141.43^\circ) + (-8)(\cos 141.43^\circ)]\mathbf{a}_\phi + 6\mathbf{a}_z \\ &= 12.81\mathbf{a}_\rho + 6\mathbf{a}_z\end{aligned}$$

(b) $\mathbf{G} = [(2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y] \cdot \mathbf{a}_\rho + [(2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y] \cdot \mathbf{a}_\phi$

Segregating the unit vectors to avoid confusion,

$$G_\rho = [(2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y] \cdot \mathbf{a}_\rho$$

Converting also the variables of rectangular to cylindrical coordinates,

$$\begin{aligned}&= (2\rho \cos \phi + \rho \sin \phi) \cos \phi - (\rho \sin \phi - 4\rho \cos \phi) \sin \phi \\ &= 2\rho \cos^2 \phi + \rho \sin \phi \cos \phi - \rho \sin^2 \phi + 4\rho \sin \phi \cos \phi \\ &= 2\rho \cos^2 \phi - \rho \sin^2 \phi + 5\rho \sin \phi \cos \phi\end{aligned}$$

$$\begin{aligned}G_\phi &= [(2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y] \cdot \mathbf{a}_\phi \\ &= (2\rho \cos \phi + \rho \sin \phi)(-\sin \phi) - (\rho \sin \phi - 4\rho \cos \phi) \cos \phi \\ &= -2\rho \sin \phi \cos \phi - \rho \sin^2 \phi - \rho \sin \phi \cos \phi + 4\rho \cos^2 \phi \\ &= -2\rho \sin \phi \cos \phi - \rho \sin^2 \phi - \rho \sin \phi \cos \phi + 4\rho \cos^2 \phi \\ &= 4\rho \cos^2 \phi - \rho \sin^2 \phi - 3\rho \sin \phi \cos \phi\end{aligned}$$

Adding together,

$$\mathbf{G} = (2\rho \cos^2 \phi - \rho \sin^2 \phi + 5\rho \sin \phi \cos \phi) \mathbf{a}_\rho + (4\rho \cos^2 \phi - \rho \sin^2 \phi - 3\rho \sin \phi \cos \phi) \mathbf{a}_\phi$$

(c) Applying the dot product from Table 3,

$$\mathbf{H} = (20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z) \cdot \mathbf{a}_x + (20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z) \cdot \mathbf{a}_y + (20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z) \cdot \mathbf{a}_z$$

$$\text{Since } \phi = (\tan^{-1}(2/5) + 180^\circ) = 201.8^\circ$$

$$H_x = 20 \cos \phi + 10 \sin \phi + 0 = 20 \cos 201.8^\circ + 10 \sin 201.8^\circ = 22.3$$

$$H_y = 20 \sin \phi - 10 \cos \phi + 0 = 20 \sin 201.8^\circ - 10 \cos 201.8^\circ = -1.86$$

$$H_z = 0 + 0 + 3 = 3$$

D1.7. Given the two points, $C(-3, 2, 1)$ and $D(r = 5, \theta = 20^\circ, \phi = -70^\circ)$, find: (a) the spherical coordinates of C ; (b) the rectangular coordinates of D ; (c) the distance from C to D .

(a) Converting rectangular to spherical coordinates,

Table 4 Rectangular to spherical coordinate systems	
$r = \sqrt{x^2 + y^2 + z^2}$	
$\theta = \cos^{-1} \frac{z}{r}$	
$\phi = \tan^{-1} y/x$	

$$r = \sqrt{(-3)^2 + 2^2 + 1^2} = 3.74$$

$$\theta = \cos^{-1}(1/3.74) = 74.5^\circ$$

$$\phi = \tan^{-1}(2/-3) + 180^\circ = 146.3^\circ$$

$$C(r = 3.74, \theta = 74.5^\circ, \phi = 146.3^\circ)$$

(b) Converting spherical to rectangular coordinates,

Table 5 Spherical to rectangular coordinate systems	
$x = r \sin \theta \cos \phi$	
$y = r \sin \theta \sin \phi$	
$z = r \cos \theta$	

$$x = 5 \sin 20^\circ \cos -70^\circ = 0.585$$

$$y = 5 \sin 20^\circ \sin -70^\circ = -1.607$$

$$z = 5 \cos 20^\circ = 4.7$$

$$D(x = 0.585, y = -1.607, z = 4.7)$$

(c) Using rectangular coordinates to find the distance

$$|\mathbf{CD}| = \sqrt{(0.585 - (-3))^2 + (-1.607 - 2)^2 + (4.7 - 1)^2} = 6.29$$

D1.8. Transform the following vectors to spherical coordinates at the points given: (a) $10\mathbf{a}_x$ at $P(x = -3, y = 2, z = 4)$; (b) $10\mathbf{a}_y$ at $Q(\rho = 5, \phi = 30^\circ, z = 4)$; (c) $10\mathbf{a}_z$ at $M(r = 4, \theta = 110^\circ, \phi = 120^\circ)$.

(a) Taking note that the dot product of the unit vectors in spherical to rectangular coordinate systems and vice versa

Table 6 Dot products of the unit vectors in spherical and rectangular coordinate systems			
	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x \cdot$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\mathbf{a}_y \cdot$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\mathbf{a}_z \cdot$	$\cos \theta$	$-\sin \theta$	0

$$10\mathbf{a}_x = 10 (\sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi)$$

From Table 4, we calculate and have $\theta = 42.03^\circ$ and $\phi = 146.31^\circ$. Substituting,

$$= 10 \sin 42.03^\circ \cos 146.31^\circ \mathbf{a}_r + 10 \cos 42.03^\circ \cos 146.31^\circ \mathbf{a}_\theta - 10 \sin 146.31^\circ \mathbf{a}_\phi$$

$$= -5.57\mathbf{a}_r - 6.18\mathbf{a}_\theta - 5.55\mathbf{a}_\phi$$

$$(b) 10\mathbf{a}_y = 10 (\sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi)$$

We manipulate to convert in cylindrical to spherical coordinates. From Table 1, we calculate and have $x = 4.33, y = 2.5$ and $z = 4$. Then from Table 4, we convert rectangular to spherical coordinates. We calculate and have $\theta = 51.34^\circ$ and $\phi = 30^\circ$. Substituting,

$$= 10 \sin 51.34^\circ \sin 30^\circ \mathbf{a}_r + 10 \cos 51.34^\circ \sin 30^\circ \mathbf{a}_\theta + 10 \cos 30^\circ \mathbf{a}_\phi$$

$$= 3.9\mathbf{a}_r - 3.12\mathbf{a}_\theta - 8.66\mathbf{a}_\phi$$

$$(c) 10\mathbf{a}_z = 10 (\cos \theta \mathbf{a}_r + (-\sin \theta) \mathbf{a}_\theta)$$

$$= 10 \cos 110^\circ \mathbf{a}_r - 10 \sin 110^\circ \mathbf{a}_\theta = -3.42\mathbf{a}_r - 9.4\mathbf{a}_\theta$$

CHAPTER 2

D2.1. A charge $Q_A = -20 \mu\text{C}$ is located at $A(-6, 4, 7)$, and a charge $Q_B = 50 \mu\text{C}$ is at $B(5, 8, -2)$ in free space. If distances are given in meters, find: (a) $\mathbf{R}_{AB}$; (b) R_{AB} . Determine the vector force exerted on Q_A by Q_B if $\epsilon_0 = (c) 10^{-9}/(36\pi) \text{ F/m}$; (d) $8.854 \times 10^{-12} \text{ F/m}$.

$$(a) \mathbf{R}_{AB} = (5 - (-6))\mathbf{a}_x + (8 - 4)\mathbf{a}_y + (-2 - 7)\mathbf{a}_z \text{ m} = 11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z \text{ m}$$

$$(b) R_{AB} = \sqrt{11^2 + 4^2 + (-9)^2} \text{ m} = 14.76 \text{ m}$$

$$(c) \mathbf{F}_{AB} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{AB}^2} \mathbf{a}_{AB} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{AB}^2} \frac{\mathbf{R}_{AB}}{R_{AB}} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{AB}^3} \mathbf{R}_{AB}$$

There is an ambiguous value of $\mathbf{F}_{AB}$ with regards from the answer of the textbook. Trying in the higher decimal number of $R_{AB} = 14.7648$,

$$= \frac{(-20 \times 10^{-6})(50 \times 10^{-6})}{4\pi(10^{-9}/36\pi)(14.7648)^3} (11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z) = (-2.7961 \times 10^{-3})(11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z)$$

$$= 30.76\mathbf{a}_x + 11.184\mathbf{a}_y - 25.16\mathbf{a}_z \text{ mN}$$

$$(d) \mathbf{F}_{AB} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{AB}^3} \mathbf{R}_{AB} = \frac{(-20 \times 10^{-6})(50 \times 10^{-6})}{4\pi(8.854 \times 10^{-12})(14.7648)^3} (11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z)$$

$$= (-2.7923 \times 10^{-3})(11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z) = 30.72\mathbf{a}_x + 11.169\mathbf{a}_y - 25.13\mathbf{a}_z \text{ mN}$$

D2.2. A charge of $-0.3 \mu\text{C}$ is located at $A(25, -30, 15)$ (in cm), and a second charge of $0.5 \mu\text{C}$ is at $B(-10, 8, 12)$ cm. Find $\mathbf{E}$ at: (a) the origin; (b) $P(15, 20, 50)$ cm.

(a) Solving first the vector and magnitude from charge A to the origin O ,

$$\mathbf{R}_{AO} = (0 - 25)\mathbf{a}_x + (0 - (-30))\mathbf{a}_y + (0 - 15)\mathbf{a}_z \text{ cm} = -25\mathbf{a}_x + 30\mathbf{a}_y - 15\mathbf{a}_z \text{ cm}$$

$$= -0.25\mathbf{a}_x + 0.3\mathbf{a}_y - 0.15\mathbf{a}_z \text{ m}$$

$$|\mathbf{R}_{AO}| = \sqrt{(-0.25)^2 + 0.3^2 + (-0.15)^2} \text{ m} = 0.4183 \text{ m}$$

$$\mathbf{E}_A = \frac{Q_A}{4\pi\epsilon_0 R_{AO}^2} \mathbf{a}_{AO} = \frac{Q_A}{4\pi\epsilon_0 R_{AO}^2} \frac{\mathbf{R}_{AO}}{R_{AO}} = \frac{Q_A}{4\pi\epsilon_0 R_{AO}^3} \mathbf{R}_{AO}$$

$$= \frac{-0.3 \times 10^{-6}}{4\pi\epsilon_0(0.4183)^3} (-0.25\mathbf{a}_x + 0.3\mathbf{a}_y - 0.15\mathbf{a}_z) \text{ V/m}$$

$$= 9.21\mathbf{a}_x - 11.05\mathbf{a}_y + 5.53\mathbf{a}_z \text{ kV/m}$$

For the charge B to the origin,

$$\mathbf{R}_{BO} = (0 - (-10))\mathbf{a}_x + (0 - 8)\mathbf{a}_y + (0 - 12)\mathbf{a}_z \text{ cm} = 10\mathbf{a}_x - 8\mathbf{a}_y - 12\mathbf{a}_z \text{ cm}$$

$$= 0.1\mathbf{a}_x - 0.08\mathbf{a}_y - 0.12\mathbf{a}_z \text{ m}$$

$$|\mathbf{R}_{BO}| = \sqrt{0.1^2 + (-0.08)^2 + (-0.12)^2} \text{ m} = 0.1755 \text{ m}$$

$$\mathbf{E}_B = \frac{Q_B}{4\pi\epsilon_0 R_{BO}^3} \mathbf{R}_{BO} = \frac{0.5 \times 10^{-6}}{4\pi\epsilon_0 (0.1755)^3} (0.1\mathbf{a}_x - 0.08\mathbf{a}_y - 0.12\mathbf{a}_z) \text{ V/m}$$

$$= 83.13\mathbf{a}_x - 66.51\mathbf{a}_y - 99.76\mathbf{a}_z \text{ kV/m}$$

Combining $\mathbf{E}_A$ and $\mathbf{E}_B$,

$$\mathbf{E} = 92.3\mathbf{a}_x - 77.6\mathbf{a}_y - 94.2\mathbf{a}_z \text{ kV/m}$$

(b) Same concept in (a)

$$\mathbf{R}_{AP} = (15 - 25)\mathbf{a}_x + (20 - (-30))\mathbf{a}_y + (50 - 15)\mathbf{a}_z \text{ cm} = -10\mathbf{a}_x + 50\mathbf{a}_y + 35\mathbf{a}_z \text{ cm}$$

$$|\mathbf{R}_{AP}| = 0.6185 \text{ m}$$

$$\mathbf{E}_A = 1.14\mathbf{a}_x - 5.7\mathbf{a}_y - 3.99\mathbf{a}_z \text{ kV/m}$$

$$\mathbf{R}_{BP} = (15 - (-10))\mathbf{a}_x + (20 - 8)\mathbf{a}_y + (50 - 12)\mathbf{a}_z \text{ cm} = 25\mathbf{a}_x + 12\mathbf{a}_y + 38\mathbf{a}_z \text{ cm}$$

$$|\mathbf{R}_{BP}| = 0.4704 \text{ m}$$

$$\mathbf{E}_B = 10.79\mathbf{a}_x + 5.18\mathbf{a}_y + 16.41\mathbf{a}_z \text{ kV/m}$$

$$\mathbf{E} = 11.9\mathbf{a}_x - 0.52\mathbf{a}_y - 12.4\mathbf{a}_z \text{ kV/m}$$

D2.3. Evaluate the sums: (a) $\sum_{m=0}^5 \frac{1+(-1)^m}{m^2+1}$; (b) $\sum_{m=1}^4 \frac{(0.1)^m+1}{(4+m^2)^{1.5}}$.

$$(a) \frac{1+(-1)^0}{0^2+1} + \frac{1+(-1)^1}{1^2+1} + \frac{1+(-1)^2}{2^2+1} + \frac{1+(-1)^3}{3^2+1} + \frac{1+(-1)^4}{4^2+1} + \frac{1+(-1)^5}{5^2+1} = 2.52$$

$$(b) \frac{(0.1)^1+1}{(4+1^2)^{1.5}} + \frac{(0.1)^2+1}{(4+2^2)^{1.5}} + \frac{(0.1)^3+1}{(4+3^2)^{1.5}} + \frac{(0.1)^4+1}{(4+4^2)^{1.5}} = 0.176$$

D2.4. Calculate the total charge within each of the indicated volumes: (a) $0.1 \leq |x|, |y|, |z| \leq 0.2$; $\rho_v = 1/(x^3y^3z^3)$; (b) $0 \leq \rho \leq 0.1$, $0 \leq \phi \leq \pi$, $2 \leq z \leq 4$; $\rho_v = \rho^2 z^2 \sin 0.6\phi$; (c) universe: $\rho_v = e^{-2r}/r^2$.

(a) The differential volume of the rectangular coordinates is $dv = dx dy dz$. Using the formula,

$$Q = \int_{vol} \rho_v dv = \int_{0.1}^{0.2} \int_{0.1}^{0.2} \int_{0.1}^{0.2} \frac{1}{x^3 y^3 z^3} dx dy dz = \int_{0.1}^{0.2} x^{-3} dx \int_{0.1}^{0.2} y^{-3} dy \int_{0.1}^{0.2} z^{-3} dz$$

$$= \left[\frac{x^{-2}}{-2} \right]_{0.1}^{0.2} \left[\frac{y^{-2}}{-2} \right]_{0.1}^{0.2} \left[\frac{z^{-2}}{-2} \right]_{0.1}^{0.2} = 112.5 \text{ C}$$

Notice that the charge Q is a large number. Neglecting the rate of change of the cubical volume, we have $xyz = (0.1)^3$ and $xyz = (0.2)^3$ which are very small; then the volume Δv is nearest to 0. Yielding,

$$Q = \rho_v \Delta v = [1/(x^3y^3z^3)] 0 = 0$$

(b) The differential volume of the cylindrical coordinates is $dv = \rho d\rho d\phi dz$. Using the formula,

$$Q = \int_0^{0.1} \int_0^\pi \int_2^4 (\rho^2 z^2 \sin 0.6\phi) \rho d\rho d\phi dz = \int_0^{0.1} \rho^3 d\rho \int_0^\pi \sin 0.6\phi d\phi \int_2^4 z^2 dz$$

$$= \left(\frac{\rho^4}{4} \right) \Big|_0^{0.1} \left(-\frac{\cos 0.6\phi}{0.6} \right) \Big|_0^\pi \left(\frac{z^3}{3} \right) \Big|_2^4 = \left(\frac{0.1^4}{4} - 0 \right) \left(\frac{-\cos 0.6(180^\circ)}{0.6} - \frac{-\cos 0.6(0^\circ)}{0.6} \right) \left(\frac{4^3}{3} - \frac{2^3}{3} \right)$$

$$= 1.018 \text{ mC}$$

(c) The differential volume of the spherical coordinates is $dv = r^2 \sin \theta d\theta d\phi dr$. Assuming this universe to be a perfect sphere, we have limits as $0 \leq r \leq \infty$, $0 \leq \phi \leq 2\pi$, $0 \leq \theta \leq \pi$. Using the formula,

$$Q = \int_0^{2\pi} \int_0^\pi \int_2^\infty (e^{-2r}/r^2) (r^2 \sin \theta d\theta d\phi dr) = \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \int_2^\infty e^{-2r} dr$$

$$= \left(\phi \right) \Big|_0^{2\pi} \left(-\cos \theta \right) \Big|_0^\pi \left(-e^{-2r}/2 \right) \Big|_2^\infty = (2\pi - 0)(-\cos 180^\circ - (-\cos 0^\circ))(-e^{-\infty}/2 - (-e^{-4}/2))$$

$$= 6.28 \text{ C}$$

D2.5. Infinite uniform line charges of 5 nC/m lie along the (positive and negative) x and y axes in free space. Find $\mathbf{E}$ at: (a) $P_A(0, 0, 4)$; (b) $P_B(0, 3, 4)$.

$$(a) \mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho \text{ where } \rho \text{ is just a radial distance } R, \text{ then}$$

$$= \frac{\rho_L}{2\pi\epsilon_0 R} \frac{\mathbf{R}}{R} = \frac{\rho_L}{2\pi\epsilon_0 R^2} \mathbf{R}$$

Finding $\mathbf{E}_x$ and $\mathbf{E}_y$ at Point A since infinite uniform line charges lie along x and y axes,

$$\mathbf{R}_x = 4\mathbf{a}_z$$

$$\mathbf{E}_x = \frac{5 \times 10^{-9}}{2\pi\epsilon_0 4^2} (4\mathbf{a}_z) = 22.469\mathbf{a}_z \text{ V/m}$$

$$\mathbf{R}_y = 4\mathbf{a}_z$$

$$\mathbf{E}_y = \frac{5 \times 10^{-9}}{2\pi\epsilon_0 4^2} (4\mathbf{a}_z) = 22.469\mathbf{a}_z \text{ V/m}$$

Adding together,

$$\mathbf{E} = 22.469\mathbf{a}_z + 22.469\mathbf{a}_z \text{ V/m} = 45\mathbf{a}_z \text{ V/m}$$

(b) Finding $\mathbf{R}$ and $\mathbf{E}$ at Point B along x and y axes,

$$\mathbf{R}_x = 3\mathbf{a}_y + 4\mathbf{a}_z$$

$$\mathbf{R}_y = 4\mathbf{a}_z$$

$$\mathbf{E}_x = \frac{5 \times 10^{-9}}{2\pi\epsilon_0 (\sqrt{3^2 + 4^2})^2} (3\mathbf{a}_y + 4\mathbf{a}_z) = 10.785\mathbf{a}_y + 14.38\mathbf{a}_z \text{ V/m}$$

$$\mathbf{E}_y = \frac{5 \times 10^{-9}}{2\pi\epsilon_0 (\sqrt{4^2})^2} (4\mathbf{a}_z) = 22.469 \mathbf{a}_z \text{ V/m}$$

$$\mathbf{E} = 10.785\mathbf{a}_y + (14.38 + 22.469)\mathbf{a}_z \text{ V/m} = 10.8\mathbf{a}_y + 36.9\mathbf{a}_z \text{ V/m}$$

D2.6. Three infinite uniform sheets of charge are located in free space as follows: 3 nC/m² at $z = -4$, 6 nC/m² at $z = 1$, and -8 nC/m² at $z = 4$. Find $\mathbf{E}$ at the point: (a) $P_A(2, 5, -5)$; (b) $P_B(4, 2, -3)$; (c) $P_C(-1, -5, 2)$; (d) $P_D(-2, 4, 5)$.

(a) Obtaining first the electric field due to charges,

$$\mathbf{E}_{z=-4} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_N = \frac{3 \times 10^{-9}}{2\epsilon_0} \mathbf{a}_z = 169.41\mathbf{a}_z \text{ V/m}$$

$$\mathbf{E}_{z=1} = \frac{6 \times 10^{-9}}{2\epsilon_0} \mathbf{a}_z = 338.82\mathbf{a}_z \text{ V/m}$$

$$\mathbf{E}_{z=4} = \frac{-8 \times 10^{-9}}{2\epsilon_0} \mathbf{a}_z = -451.76\mathbf{a}_z \text{ V/m}$$

Since the point A is located below all the sheets of charge, all directions of electric field $\mathbf{E}$ are negative $\mathbf{a}_z$ direction with relative to normal, which is $\mathbf{a}_N$, of P_A . Combining together,

$$\mathbf{E} = -\mathbf{E}_1 - \mathbf{E}_2 - \mathbf{E}_3 = -169.41\mathbf{a}_z - 338.82\mathbf{a}_z - (-451.76\mathbf{a}_z) \text{ V/m} = -56.5\mathbf{a}_z \text{ V/m}$$

(b) The location of point B suggests that the electric field $\mathbf{E}$ contributed by the surface charge at $z = -4$ will be in positive $\mathbf{a}_z$ direction while others are negative. Combining together,

$$\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2 - \mathbf{E}_3 = 282.3\mathbf{a}_z \text{ V/m}$$

(c) Same arguments in (b), only $z = 4$ will be in negative $\mathbf{a}_z$ direction. Combining together,

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 - \mathbf{E}_3 = 960\mathbf{a}_z \text{ V/m}$$

(d) We know that the point D is located above all the sheets of charge and all electric fields $\mathbf{E}$ are positive $\mathbf{a}_z$ direction. Combining together,

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 = 56.5\mathbf{a}_z \text{ V/m}$$

D2.7. Find the equation of that streamline that passes through the point $P(1, 4, -2)$ in the field

$$\mathbf{E} = (a) \frac{-8x}{y} \mathbf{a}_x + \frac{4x^2}{y^2} \mathbf{a}_y; (b) 2e^{5x}[y(5x+1)\mathbf{a}_x + x\mathbf{a}_y].$$

$$(a) \frac{dy}{dx} = \frac{\mathbf{E}_y}{\mathbf{E}_x} = \frac{4x^2/y^2}{-8x/y} = -\frac{x}{2y}$$

$$2ydy = -xdx$$

$$\int 2ydy = \int -xdx$$

$$y^2 = -\frac{x^2}{2} + \frac{C^2}{2}$$

For $P(x = 1, y = 4, z = -2)$,

$$4^2 = -\frac{1^2}{2} + \frac{C^2}{2}; C^2 = 33$$

$$x^2 + 2y^2 = 33$$

$$(b) \frac{dy}{dx} = \frac{x}{y(5x+1)}$$

$$\int ydy = \int \frac{x}{5x+1} dx$$

Letting $u = 5x + 1$ and $du = 5 dx$, and then $x = (u - 1)/5$,

$$\frac{y^2}{2} = \int \frac{u-1}{5} \left(\frac{1}{u} \right) \left(\frac{du}{5} \right) = \int \frac{u-1}{25u} du = \int \frac{1}{25} du - \int \frac{1}{25u} du = \frac{1}{25}(u - \ln|u|)$$

$$\frac{y^2}{2} = \frac{1}{25}[(5x+1) - \ln|5x+1|] + \frac{C^2}{2}$$

For $P(x = 1, y = 4, z = -2)$,

$$\frac{4^2}{2} = \frac{1}{25}[(5(1)+1) - \ln|5(1)+1|] + \frac{C^2}{2}; C^2 = 15.66$$

$$\frac{y^2}{2} = 0.04(5x+1) - 0.04 \ln|5x+1| + \frac{15.66}{2}$$

$$y^2 = 15.7 + 0.4x - 0.08 \ln|5x+1|$$

CHAPTER 3

D3.1. Given a 60- μC point charge located at the origin, find the total electric flux passing through: (a) that portion of the sphere $r = 26$ cm bounded by $0 < \theta < \pi/2$ and $0 < \phi < \pi/2$; (b) the closed surface defined by $\rho = 26$ cm and $z = \pm 26$ cm; (c) the plane $z = 26$ cm.

$$(a) \mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$$

$$d\Psi = \frac{Q}{4\pi r^2} \mathbf{a}_r (r^2 \sin \theta d\theta d\phi) \mathbf{a}_r = \frac{Q}{4\pi} \sin \theta d\theta d\phi$$

$$\Psi = \frac{Q}{4\pi} \int_0^{\pi/2} \int_0^{\pi/2} \sin \theta d\theta d\phi = \frac{Q}{4\pi} \int_0^{\pi/2} \sin \theta d\theta \int_0^{\pi/2} d\phi = \frac{Q}{4\pi} (-\cos \theta|_0^{\pi/2}) (\phi|_0^{\pi/2})$$

$$= \frac{60 \times 10^{-6}}{4\pi} (-\cos 90^\circ - (-\cos 0^\circ)) ((\pi/2) - 0) = 7.5 \mu\text{C}$$

(b) Deriving Gauss's law of the cylinder,

$$Q = \oint \mathbf{D}_s \cdot d\mathbf{S} = D_s \oint \rho d\phi dz = D_s \rho \int_0^\pi d\phi \int_0^L dz = D_s \rho \phi|_0^\pi z|_0^L = D_s 2\pi \rho L$$

$$D_s = D_\rho = \frac{Q}{2\pi \rho L} \quad \text{or} \quad \mathbf{D}_\rho = \frac{Q}{2\pi \rho L} \mathbf{a}_\rho$$

Solving Ψ ,

$$d\Psi = \frac{Q}{2\pi \rho L} \mathbf{a}_\rho (\rho d\phi dz) \mathbf{a}_\rho = \frac{Q}{2\pi \rho L} \rho d\phi dz = \frac{Q}{2\pi \rho L} \rho \int_0^{2\pi} d\phi \int_0^L dz = \frac{Q}{2\pi \rho L} (2\pi \rho L)$$

$$\Psi = Q = 60 \mu\text{C}$$

(c) Getting the equation in (b) with changing the limits of ϕ as $0 \leq \phi \leq \pi$ because only the flux lines pass in half through the plane,

$$d\Psi = \frac{Q}{2\pi \rho L} \rho \int_0^\pi d\phi \int_0^L dz = \frac{Q}{2\pi \rho L} (\pi \rho L)$$

$$\Psi = Q/2 = 30 \mu\text{C}$$

D3.2. Calculate **D** in rectangular coordinates at point $P(2, -3, 6)$ produced by: (a) a point charge $Q_A = 55 \text{ mC}$ at $Q(-2, 3, -6)$; (b) a uniform line charge $\rho_{LB} = 20 \text{ mC/m}$ on the axis; (c) a uniform surface charge density $\rho_{SC} = 120 \text{ } \mu\text{C/m}^2$ on the plane $z = -5 \text{ m}$.

$$(a) \mathbf{D} = \epsilon_0 \mathbf{E} = \frac{Q}{4\pi R^2} \mathbf{a}_R = \frac{Q}{4\pi R^3} \mathbf{R}$$

$$\mathbf{R}_{QP} = (2 - (-2))\mathbf{a}_x + (-3 - 3)\mathbf{a}_y + (6 - (-6))\mathbf{a}_z = 4\mathbf{a}_x - 6\mathbf{a}_y + 12\mathbf{a}_z$$

$$R_{QP} = 14$$

$$\mathbf{D}_{QP} = \frac{Q}{4\pi R_{QP}^3} \mathbf{R}_{QP} = \frac{55 \times 10^{-3}}{4\pi 14^3} (4\mathbf{a}_x - 6\mathbf{a}_y + 12\mathbf{a}_z) = 6.38\mathbf{a}_x - 9.57\mathbf{a}_y + 19.14\mathbf{a}_z \text{ } \mu\text{C/m}^2$$

$$(b) \mathbf{D} = \epsilon_0 \mathbf{E} = \frac{\rho_L}{2\pi R} \mathbf{a}_R = \frac{\rho_L}{2\pi R^2} \mathbf{R}$$

$$\mathbf{R}_x = (2 - x)\mathbf{a}_x - 3\mathbf{a}_y + 6\mathbf{a}_z$$

Since the infinite line charge density is along x axis, the electric field E at point P is having only y and z components present. Canceling x component due to symmetry,

$$\mathbf{R}_x = -3\mathbf{a}_y + 6\mathbf{a}_z$$

$$R_x = \sqrt{45}$$

$$\mathbf{D}_x = \frac{\rho_L}{2\pi R_x^2} \mathbf{R}_x = \frac{20 \times 10^{-3}}{2\pi (\sqrt{45})^2} (-3\mathbf{a}_y + 6\mathbf{a}_z) = -212\mathbf{a}_y + 424\mathbf{a}_z \text{ } \mu\text{C/m}^2$$

$$(c) \mathbf{D} = \epsilon_0 \mathbf{E} = \frac{\rho_s}{2} \mathbf{a}_R$$

The infinite surface charge density is an infinite x - y plane located at $z = -5$ and the charge is spread on that plane. Going back the formula,

$$\mathbf{D} = (\rho_s/2)\mathbf{a}_z = (120/2) \mu\text{C/m}^2 = 60\mathbf{a}_z \text{ } \mu\text{C/m}^2$$

D3.3. Given the electric flux density, $\mathbf{D} = 0.3r^2\mathbf{a}_r \text{ nC/m}^2$ in free space: (a) find $\mathbf{E}$ at point $P(r = 2, \theta = 25^\circ, \phi = 90^\circ)$; (b) find the total charge within the sphere $r = 3$; (c) find the total electric flux leaving the sphere $r = 4$.

$$(a) \mathbf{E} = \mathbf{D}/\epsilon_0 = (0.3 \times 10^{-9} r^2 \mathbf{a}_r)/\epsilon_0 = (0.3 \times 10^{-9} (2^2) \mathbf{a}_r)/\epsilon_0 = 135.5\mathbf{a}_r \text{ V/m}$$

$$(b) Q = \oint \mathbf{D} \cdot d\mathbf{S} = 0.3 \times 10^{-9} r^2 \mathbf{a}_r \int_0^{2\pi} \int_0^\pi r^2 \sin \theta d\theta d\phi \mathbf{a}_r = 0.3 \times 10^{-9} r^2 (4\pi r^2) \\ = 0.3 \times 10^{-9} (4\pi r^4) = 0.3 \times 10^{-9} (4\pi (3^4)) = 305 \text{ nC}$$

(c) Same procedure in (b),

$$\Psi = Q = 1.2 \times 10^{-9} \pi r^4 = 1.2 \times 10^{-9} \pi (4)^4 = 965 \text{ nC}$$

D3.4. Calculate the total electric flux leaving the cubical surface formed by the six planes $x, y, z = \pm 5$ if the charge distribution is: (a) two point charges, $0.1 \text{ } \mu\text{C}$ at $(1, -2, 3)$ and $1/7 \text{ } \mu\text{C}$ at $(-1, 2, -2)$; (b) a uniform line charge of $\pi \text{ } \mu\text{C/m}$ at $x = -2, y = 3$; (c) a uniform surface charge of $0.1 \text{ } \mu\text{C/m}^2$ on the plane $y = 3x$.

(a) Both given charges are enclosed by the cubical volume according to Gauss's law. Adding,

$$\Psi = Q_1 + Q_2 = 0.1 \text{ } \mu\text{C} + (1/7) \text{ } \mu\text{C} = 0.243 \text{ } \mu\text{C}$$

(b) The line charge distribution passes through $x = -2$ and $y = 3$; it is parallel to z axis. The total length of that charge distribution enclosed by the given cubical volume is 10 units as $z = \pm 5$. Applying Gauss's law of line charge,

$$\Psi = \rho_L L = \pi (10) \text{ } \mu\text{C} = 31.4 \text{ } \mu\text{C}$$

(c) A straight line equation, $y = 3x$, passes through the origin. We need to find the length of that line which is enclosed by the given volume. The length is moving up and down along z axis as $z = \pm 5$ to form a plane. Putting $y = 5$,

$$5 = 3x; x = 5/3$$

Finding the length of that line on the plane formed by positive x and y axes,

$$\sqrt{5^2 + (5/3)^2} = 5.27$$

The same length we get on the plane formed by negative x and y axes. Adding two lengths, $5.27 + 5.27 = 10.54$

That straight line moves between $z = \pm 5$ to form a plane. Knowing the area,

$$10(10.54) = 105.4$$

Applying Gauss's law of surface charge,

$$\Psi = \rho_S S = (0.1 \mu\text{C})105.4 = 10.54 \mu\text{C}$$

D3.5. A point charge of $0.25 \mu\text{C}$ is located at $r = 0$, and uniform surface charge densities are located as follows: 2 mC/m^2 at $r = 1 \text{ cm}$, and -0.6 mC/m^2 at $r = 1.8 \text{ cm}$. Calculate $\mathbf{D}$ at: (a) $r = 0.5 \text{ cm}$; (b) $r = 1.5 \text{ cm}$; (c) $r = 2.5 \text{ cm}$. (d) What uniform charge density should be established at $r = 3 \text{ cm}$ to cause $\mathbf{D} = 0$ at $r = 3.5 \text{ cm}$?

Note: Always remember the radius r within the charge Q and charge density D are different. Assume them as $r_{(Q)}$ and $r_{(D)}$.

$$(a) \mathbf{D} = \frac{Q_1}{4\pi r_{(D)}^2} \mathbf{a}_r = \frac{0.25 \times 10^{-6}}{4\pi(0.5 \times 10^{-2})^2} \mathbf{a}_r = 796 \mathbf{a}_r \mu\text{C}$$

(b) Finding the second charge Q_2 ,

$$Q_2 = \rho_S S = \rho_S (4\pi r_{(Q)}^2) = 2 \times 10^{-3} (4\pi(1 \times 10^{-2})^2) = 2.513 \times 10^{-6} \text{ C}$$

$$\mathbf{D} = \frac{Q_1 + Q_2}{4\pi r_{(D)}^2} \mathbf{a}_r = \frac{0.25 \times 10^{-6} + 2.513 \times 10^{-6}}{4\pi(1.5 \times 10^{-2})^2} \mathbf{a}_r = 977 \mathbf{a}_r \mu\text{C}$$

(c) Finding the third charge Q_3 ,

$$Q_3 = \rho_S (4\pi r_{(Q)}^2) = -0.6 \times 10^{-3} (4\pi(1.8 \times 10^{-2})^2) = -2.443 \times 10^{-6} \text{ C}$$

$$\mathbf{D} = \frac{Q_1 + Q_2 + Q_3}{4\pi r_{(D)}^2} \mathbf{a}_r = \frac{0.25 \times 10^{-6} + 2.513 \times 10^{-6} - 2.443 \times 10^{-6}}{4\pi(2.5 \times 10^{-2})^2} \mathbf{a}_r = 40.8 \mathbf{a}_r \mu\text{C}$$

$$(d) 0 = \mathbf{D} = \frac{Q_1 + Q_2 + Q_3 + Q_4}{4\pi r_{(D)}^2} \mathbf{a}_r = \frac{Q_1 + Q_2 + Q_3 + \rho_S S}{4\pi r_{(D)}^2} \mathbf{a}_r = \frac{0.32 \times 10^{-6} + \rho_S 4\pi r_{(Q)}^2}{4\pi r_{(D)}^2} \mathbf{a}_r$$

$$0 = (0.32 \times 10^{-6} + 4\rho_S \pi r_{(Q)}^2) \mathbf{a}_r$$

$$-0.32 \times 10^{-6} = 4\rho_S \pi r_{(Q)}^2$$

$$\rho_S = -\frac{0.32 \times 10^{-6}}{4\pi r_{(Q)}^2} = -\frac{0.32 \times 10^{-6}}{4\pi(3 \times 10^{-2})^2} = -28.3 \mu\text{C/m}^2$$

D3.6. In free space, let $\mathbf{D} = 8xyz^4 \mathbf{a}_x + 4x^2z^4 \mathbf{a}_y + 16x^2yz^3 \mathbf{a}_z \text{ pC/m}^2$. (a) Find the total electric flux passing through the rectangular surface $z = 2$, $0 < x < 2$, $1 < y < 3$, in the $\mathbf{a}_z$ direction. (b) Find $\mathbf{E}$ at $P(2, -1, 3)$. (c) Find an approximate value for the total charge contained in an incremental sphere located at $P(2, -1, 3)$ and having a volume of 10^{-12} m^3 .

$$(a) \Psi = \oint \mathbf{D} \cdot d\mathbf{S} \text{ where } d\mathbf{S} = dx dy \mathbf{a}_z$$

$$= \int_1^3 \int_0^2 (8xyz^4 \mathbf{a}_x + 4x^2z^4 \mathbf{a}_y + 16x^2yz^3 \mathbf{a}_z) \cdot dx dy \mathbf{a}_z = \int_1^3 \int_0^2 16x^2yz^3 dx dy$$

$$= z^3 \int_1^3 y dy \int_0^2 16x^2 dx = z^3 (y^2/2) \Big|_1^3 (16x^3/3) \Big|_0^2 = [16z^3(4)(8/3)]_{z=2} = 1365 \text{ pC}$$

$$(b) \mathbf{E} = \mathbf{D}/\epsilon_0 = [(8xyz^4 \mathbf{a}_x + 4x^2z^4 \mathbf{a}_y + 16x^2yz^3 \mathbf{a}_z) \times 10^{-12}]/\epsilon_0 \Big|_{x=2, y=-1, z=3} \\ = -146.4 \mathbf{a}_x + 146.4 \mathbf{a}_y - 195.2 \mathbf{a}_z \text{ V/m}$$

$$(c) Q = \text{Charge enclosed in volume } \Delta v \doteq \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \times \Delta v$$

Finding the partial derivatives of each term,

$$\partial D_x / \partial x = \partial(8xyz^4) / \partial x = (8yz^4)_{x=2, y=-1, z=3} = -648$$

$$\partial D_y / \partial y = \partial(4x^2z^4) / \partial y = 0$$

$$\partial D_z / \partial z = \partial(16x^2yz^3) / \partial z = (48x^2yz^2)_{x=2, y=-1, z=3} = -1728$$

$$Q = (-648 - 1728) \times 10^{-12} \Delta v = [(-648 - 1728) \times 10^{-12}] (10^{-12}) = -2.38 \times 10^{-21} \text{ C}$$

D3.7. In each of the following parts, find a numerical value for $\text{div } \mathbf{D}$ at the point specified: (a) $\mathbf{D} = (2xyz - y^2) \mathbf{a}_x + (x^2z - 2xy) \mathbf{a}_y + x^2y \mathbf{a}_z \text{ C/m}^2$ at $P_A(2, 3, -1)$; (b) $\mathbf{D} = 2\rho z^2 \sin^2 \phi \mathbf{a}_\rho + \rho z^2 \sin 2\phi \mathbf{a}_\phi + 2\rho^2 z \sin^2 \phi \mathbf{a}_z \text{ C/m}^2$ at $P_B(\rho = 2, \phi = 110^\circ, z = -1)$; (c) $\mathbf{D} = 2r \sin \theta \cos \phi \mathbf{a}_r + r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi \text{ C/m}^2$ at $P_C(r = 1.5, \theta = 30^\circ, \phi = 50^\circ)$.

$$(a) \text{div } \mathbf{D} = \partial D_x / \partial x + \partial D_y / \partial y + \partial D_z / \partial z$$

Finding the partial derivatives of each term,

$$\partial D_x / \partial x = \partial(2xyz - y^2) / \partial x = 2yz$$

$$\partial D_y / \partial y = \partial(x^2 z - 2xy) / \partial y = -2x$$

$$\partial D_z / \partial z = \partial(x^2 y) / \partial z = 0$$

$$\text{div } \mathbf{D} = (2yz - 2x)_{x=2, y=3, z=-1} = -10$$

$$(b) \text{div } \mathbf{D} = \frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Finding the partial derivatives of each term,

$$\frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} = \frac{1}{\rho} \frac{\partial(2\rho^2 z^2 \sin^2 \phi)}{\partial \rho} = (4z^2 \sin^2 \phi)_{\rho=2, \phi=110^\circ, z=-1} = 3.532$$

$$\frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} = \frac{1}{\rho} \frac{\partial(\rho z^2 \sin 2\phi)}{\partial \phi} = (2z^2 \cos 2\phi)_{\rho=2, \phi=110^\circ, z=-1} = -1.532$$

$$\frac{\partial D_z}{\partial z} = \frac{\partial(2\rho^2 z \sin^2 \phi)}{\partial z} = (2\rho^2 \sin^2 \phi)_{\rho=2, \phi=110^\circ, z=-1} = 7.064$$

$$\text{div } \mathbf{D} = 3.532 + (-1.532) + 7.064 = 9.06$$

$$(c) \text{div } \mathbf{D} = \frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta D_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

$$\frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} = \frac{1}{r^2} \frac{\partial(2r^3 \sin \theta \cos \phi)}{\partial r} = (6 \sin \theta \cos \phi)_{r=1.5, \theta=30^\circ, \phi=50^\circ} = 1.928$$

$$\frac{1}{r \sin \theta} \frac{\partial(\sin \theta D_\theta)}{\partial \theta} = \frac{1}{r \sin \theta} \frac{\partial(r \sin \theta \cos \theta \cos \phi D_\theta)}{\partial \theta}$$

Applying product rule of the derivative of $\sin \theta \cos \theta$,

$$= \frac{\cos \phi}{\sin \theta} (\cos^2 \theta - \sin^2 \theta)_{r=1.5, \theta=30^\circ, \phi=50^\circ} = 0.643$$

$$\frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi} = \frac{1}{r \sin \theta} \frac{\partial(-r \sin \phi)}{\partial \phi} = (-\cos \phi / \sin \theta)_{r=1.5, \theta=30^\circ, \phi=50^\circ} = -1.286$$

$$\text{div } \mathbf{D} = 1.928 + 0.643 - 1.286 = 1.29$$

D3.8. Determine an expression for the volume charge density associated with each $\mathbf{D}$ field: (a)

$\mathbf{D} = \frac{4xy}{z} \mathbf{a}_x + \frac{2x^2}{z} \mathbf{a}_y - \frac{2x^2 y}{z^2} \mathbf{a}_z$; (b) $\mathbf{D} = z \sin \phi \mathbf{a}_\rho + z \cos \phi \mathbf{a}_\phi + \rho \sin \phi \mathbf{a}_z$; (c) $\mathbf{D} = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$.

Note: The volume charge density ρ_v is equal to $\text{div } \mathbf{D}$.

$$(a) \partial D_x / \partial x = \partial(4xy/z) / \partial x = 4y/z$$

$$\partial D_y / \partial y = \partial(2x^2/z) / \partial y = 0$$

$$\partial D_z / \partial z = \partial(-2x^2 y / z^2) / \partial z = 4x^2 y / z^3$$

$$\rho_v = 4y/z + 4x^2 y / z^3 = (4y/z^3)(x^2 + z^2)$$

$$(b) (1/\rho) \partial(\rho D_\rho) / \partial \rho = (1/\rho) \partial(\rho z \sin \phi) / \partial \rho = (z \sin \phi) / \rho$$

$$(1/\rho) \partial D_\phi / \partial \phi = (1/\rho) \partial(z \cos \phi) / \partial \phi = (-z \sin \phi) / \rho$$

$$\partial D_z / \partial z = \partial(\rho \sin \phi) / \partial z = 0$$

$$\rho_v = (z \sin \phi) / \rho + (-z \sin \phi) / \rho = 0$$

$$(c) (1/r^2) \partial(r^2 D_r) / \partial r = (1/r^2) \partial(r^2 \sin \theta \sin \phi) / \partial r = (2 \sin \theta \sin \phi) / r$$

$$(1/r \sin \theta) \partial(\sin \theta D_\theta) / \partial \theta = (1/r \sin \theta) \partial(\sin \theta \cos \theta \sin \phi) / \partial \theta$$

$$= (\sin \phi / r \sin \theta) (\cos^2 \theta - \sin^2 \theta)$$

$$(1/r \sin \theta) \partial D_\phi / \partial \phi = (1/r \sin \theta) \partial(\cos \phi) / \partial \phi = -\sin \phi / (r \sin \theta)$$

$$\rho_v = (2 \sin \theta \sin \phi) / r + (\sin \phi / r \sin \theta) (\cos^2 \theta - \sin^2 \theta) + (-\sin \phi / (r \sin \theta))$$

$$= \frac{2 \sin^2 \theta \sin \phi + \cos^2 \theta \sin \phi - \sin^2 \theta \sin \phi - \sin \phi}{r \sin \theta} = \frac{\sin^2 \theta \sin \phi + \cos^2 \theta \sin \phi - \sin \phi}{r \sin \theta}$$

Since $\sin^2 \theta + \cos^2 \theta = 1$,

$$= \frac{\sin \phi (\sin^2 \theta + \cos^2 \theta) - \sin \phi}{r \sin \theta} = \frac{0}{r \sin \theta} = 0$$

D3.9. Given the field $\mathbf{D} = 6\rho \sin(\phi/2) \mathbf{a}_\rho + 1.5\rho \cos(\phi/2) \mathbf{a}_\phi$ C/m², evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\phi = 0$, $\phi = \pi$, $z = 0$, and $z = 5$.

Using the divergence theorem with the formula: $\oint_S \mathbf{D} \cdot d\mathbf{S}$,

$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{S} &= \oint_S (6\rho \sin(\phi/2) \mathbf{a}_\rho + 1.5\rho \cos(\phi/2) \mathbf{a}_\phi) \cdot (\rho d\phi dz \mathbf{a}_\rho - d\rho dz \mathbf{a}_\phi) \\&= \oint_S 6\rho \sin(\phi/2) \rho d\phi dz - \oint_S 1.5\rho \cos(\phi/2) d\rho dz \\&= \int_0^\pi \int_0^5 6\rho \sin(\phi/2) \rho d\phi dz - \int_0^2 \int_0^5 1.5\rho \cos(\phi/2) d\rho dz \\&= 6\rho^2 \int_0^\pi \sin(\phi/2) d\phi \int_0^5 dz - (1.5 \cos(\phi/2)) \int_0^2 \rho d\rho \int_0^5 dz \\&= 24 \left(-2 \cos \phi/2 \Big|_0^\pi \right) \left(z \Big|_0^5 \right) - (1.5 \cos \phi/2) \left(\rho^2/2 \Big|_0^2 \right) \left(z \Big|_0^5 \right)\end{aligned}$$

Since the second surface lies at $\phi = 0^\circ$,

$$= 24(2)(5) - (1.5 \cos \phi/2)(2)(5) = 240 - 15 = 225$$

Using another divergence theorem with the formula: $\int_{\text{vol}} \nabla \cdot \mathbf{D} dv$ but solving first the $\nabla \cdot \mathbf{D}$,

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \text{div } \mathbf{D} = (1/\rho) \partial(\rho D_\rho)/\partial\rho + (1/\rho) \partial D_\phi/\partial\phi + \partial D_z/\partial z \\&= (1/\rho) \partial(6\rho^2 \sin \phi/2)/\partial\rho + (1/\rho) \partial(1.5\rho \cos \phi/2)/\partial\phi + 0 \\&= 12 \sin \phi/2 + (-1.5/2) \sin \phi/2 = 11.25 \sin \phi/2\end{aligned}$$

$$\begin{aligned}\int_{\text{vol}} \nabla \cdot \mathbf{D} dv &= \int_0^2 \int_0^\pi \int_0^5 (11.25 \sin \phi/2) \rho d\rho d\phi dz = 11.25 \int_0^2 \rho d\rho \int_0^\pi (\sin \phi/2) d\phi \int_0^5 dz \\&= 11.25 \left(\rho^2/2 \Big|_0^2 \right) \left(-2 \cos \phi/2 \Big|_0^\pi \right) \left(z \Big|_0^5 \right) = 11.25(2)(2)(5) = 225\end{aligned}$$

CHAPTER 4

D4.1. Given the electric field $\mathbf{E} = (1/z^2)(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z)$ V/m, find the differential amount of work done in moving a 6-nC charge a distance of 2 μm , starting at $P(2, -2, 3)$ and proceeding in the direction $\mathbf{a}_L = (a) -6/7\mathbf{a}_x + 3/7\mathbf{a}_y + 2/7\mathbf{a}_z$; (b) $6/7\mathbf{a}_x - 3/7\mathbf{a}_y - 2/7\mathbf{a}_z$; (c) $3/7\mathbf{a}_x + 6/7\mathbf{a}_y$.

(a) $dW = -Q\mathbf{E} \cdot d\mathbf{L}$

Finding first the differential length $d\mathbf{L}$,

$$\begin{aligned}d\mathbf{L} &= \mathbf{a}_L \cdot dL = (-6/7\mathbf{a}_x + 3/7\mathbf{a}_y + 2/7\mathbf{a}_z)(2 \times 10^{-6}) = (-12/7\mathbf{a}_x + 6/7\mathbf{a}_y + 4/7\mathbf{a}_z)(1 \times 10^{-6}) \\dW &= -(6 \times 10^{-9})(1/z^2)(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z) \cdot (-12/7\mathbf{a}_x + 6/7\mathbf{a}_y + 4/7\mathbf{a}_z)(1 \times 10^{-6}) \\&= -6 \times 10^{-15}[(1/z^2)((-96/7)xyz + (24/7)x^2z - (16/7)x^2y)]_{x=2, y=-2, z=3} = -6 \times 10^{-15}(224/9) \\&= -149.3 \text{ fJ}\end{aligned}$$

(b) Same procedure in (a),

$$\begin{aligned}d\mathbf{L} &= (12/7\mathbf{a}_x - 6/7\mathbf{a}_y + 4/7\mathbf{a}_z)(1 \times 10^{-6}) \\dW &= -(6 \times 10^{-9})(1/z^2)(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z) \cdot (12/7\mathbf{a}_x - 6/7\mathbf{a}_y + 4/7\mathbf{a}_z)(1 \times 10^{-6}) \\&= -6 \times 10^{-15}[(1/z^2)((96/7)xyz - (24/7)x^2z - (16/7)x^2y)]_{x=2, y=-2, z=3} = -6 \times 10^{-15}(-224/9) \\&= 149.3 \text{ fJ}\end{aligned}$$

(c) Same procedure in (a) and (b),

$$\begin{aligned}d\mathbf{L} &= (6/7\mathbf{a}_x + 12/7\mathbf{a}_y + 0\mathbf{a}_z)(1 \times 10^{-6}) \\dW &= -(6 \times 10^{-9})(1/z^2)(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z) \cdot (6/7\mathbf{a}_x + 12/7\mathbf{a}_y + 0\mathbf{a}_z)(1 \times 10^{-6}) \\&= -6 \times 10^{-15}[(1/z^2)((48/7)xyz + (48/7)x^2z)]_{x=2, y=-2, z=3} = -6 \times 10^{-15}(0) \\&= 0\end{aligned}$$

D4.2. Calculate the work done in moving a 4-C charge from $B(1, 0, 0)$ to $A(0, 2, 0)$ along the path $y = 2 - 2x$, $z = 0$ in the field $\mathbf{E} = (a) 5\mathbf{a}_x$ V/m; (b) $5x\mathbf{a}_x$ V/m; (c) $5x\mathbf{a}_x + 5y\mathbf{a}_y$ V/m.

$$\begin{aligned}(a) W &= -Q \int_B^A \mathbf{E} \cdot d\mathbf{L} \text{ where } d\mathbf{L} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z \\&= -4 \int_1^0 (5\mathbf{a}_x + 0\mathbf{a}_y + 0\mathbf{a}_z) \cdot (dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z) = -4 \int_1^0 5 dx = -20x \Big|_1^0 = -20(-1) = 20 \text{ J}\end{aligned}$$

(b) Same procedure in (a),

$$W = -4 \int_1^0 5x dx = -20(x^2/2) \Big|_1^0 = -20(-0.5) = 10 \text{ J}$$

(c) Same procedure in (a) and (b),

$$W = -4 \int_{(1,0)}^{(0,2)} (5x dx + 5y dy) = -20 \left(\frac{x^2}{2} + \frac{y^2}{2} \right) \Big|_{(1,0)}^{(0,2)} = -20(1.5) = -30 \text{ J}$$

D4.3. We will see later that a time-varying $\mathbf{E}$ field need not be conservative. (If it is not conservative, the work expressed by equation: $W = -Q \int_{\text{initial}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$ may be a function of the path used.) Let $\mathbf{E} = y\mathbf{a}_x$ V/m at a certain instant of time, and calculate the work required to move a 3-C charge from (1, 3, 5) to (2, 0, 3) along the straight-line segments joining: (a) (1, 3, 5) to (2, 3, 5) to (2, 0, 5) to (2, 0, 3); (b) (1, 2, 5) to (1, 3, 3) to (1, 0, 3) to (2, 0, 3).

(a) Use the formula in D4.2 (a). One differential unit vector of each line segments is used. For segment (1, 3, 5) to (2, 3, 5),

$$W_1 = -3 \int_1^2 (y\mathbf{a}_x + 0\mathbf{a}_y + 0\mathbf{a}_z) \cdot dx\mathbf{a}_x = -3y \int_1^2 dx = -3y \left(x \Big|_1^2 \right)_{y=3} = -9 \text{ J}$$

For segment (2, 3, 5) to (2, 0, 5),

$$W_2 = -3 \int_1^2 (y\mathbf{a}_x + 0\mathbf{a}_y + 0\mathbf{a}_z) \cdot dy\mathbf{a}_y = 0$$

For segment (2, 0, 5) to (2, 0, 3),

$$W_3 = -3 \int_1^2 (y\mathbf{a}_x + 0\mathbf{a}_y + 0\mathbf{a}_z) \cdot dz\mathbf{a}_z = 0$$

Adding the total work, $W = W_1 + W_2 + W_3 = -9 \text{ J}$

(b) Same procedure in (a). For segment (1, 2, 5) to (1, 3, 3),

$$W_1 = -3y \int_1^1 dx = -3y \left(x \Big|_1^1 \right)_{y=3} = 0$$

Both W_2 and W_3 are zeroes no matter what the line segments are. So adding the total work, $W = 0$.

D4.4. An electric field is expressed in rectangular coordinates by $\mathbf{E} = 6x^2\mathbf{a}_x + 6y\mathbf{a}_y + 4z\mathbf{a}_z$ V/m. Find: (a) V_{MN} if points M and N are specified by $M(2, 6, -1)$ and $N(-3, -3, 2)$; (b) V_M if $V = 0$ at $Q(4, -2, -35)$; (c) V_N if $V = 2$ at $P(1, 2, -4)$.

$$\begin{aligned} (a) V_{MN} &= -\int_N^M \mathbf{E} \cdot d\mathbf{L} = -\int_{(-3,-3,2)}^{(2,6,-1)} (6x^2 dx + 6y dy + 4dz) = -(2x^2 + 3y^2 + 4z) \Big|_{(-3,-3,2)}^{(2,6,-1)} \\ &= -(70 + 81 - 12) = -139 \text{ V} \end{aligned}$$

$$(b) V_{MQ} = -\int_Q^M \mathbf{E} \cdot d\mathbf{L} = -(2x^2 + 3y^2 + 4z) \Big|_{(4,-2,-35)}^{(2,6,-1)} = -(-112 + 96 + 136) = -120 \text{ V}$$

$$V_{MQ} = V_M - V_Q$$

$$V_M = V_{MQ} + V_Q = -120 \text{ V} + 0 = -120 \text{ V}$$

$$(c) V_{NP} = -\int_P^N \mathbf{E} \cdot d\mathbf{L} = -(2x^2 + 3y^2 + 4z) \Big|_{(1,2,-4)}^{(-3,-3,2)} = -(-56 + 15 + 24) = 17 \text{ V}$$

$$V_{NP} = V_N - V_Q$$

$$V_N = V_{NP} + V_P = 17 \text{ V} + 2 \text{ V} = 19 \text{ V}$$

D4.5. A 15-nC point charge is at the origin in free space. Calculate V_1 if point P_1 is located at $P_1(-2, 3, -1)$ and (a) $V = 0$ at (6, 5, 4); (b) $V = 0$ at infinity; (c) $V = 5 \text{ V}$ at (2, 0, 4).

(a) Assume that second point is M . Applying the potential difference V_{AB} , or V_{PM} in the problem,

$$V_{PM} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_P} - \frac{1}{R_M} \right)$$

$$R_P = \sqrt{(-2)^2 + 3^2 + (-1)^2} = \sqrt{14}$$

$$R_M = \sqrt{6^2 + 5^2 + 4^2} = \sqrt{77}$$

$$V_{PM} = \frac{15 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{14}} - \frac{1}{\sqrt{77}} \right) = 20.67 \text{ V}$$

$$V_{PM} = V_P - V_M$$

$$V_P = V_{PM} + V_M = 20.67 \text{ V} + 0 = 20.67 \text{ V}$$

(b) Same procedure in (a),

$$R_P = \sqrt{14}$$

$$R_M = \infty$$

$$V_{PM} = \frac{15 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{14}} - \frac{1}{\infty} \right) = 36 \text{ V}$$

$$V_P = V_{PM} + V_M = 36 \text{ V} + 0 = 36 \text{ V}$$

(c) Same procedure in (a) and (b),

$$R_P = \sqrt{14}$$

$$R_M = \sqrt{2^2 + 4^2} = 20$$

$$V_{PM} = \frac{15 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{14}} - \frac{1}{\sqrt{20}} \right) = 5.89 \text{ V}$$

$$V_P = V_{PM} + V_M = 5.89 \text{ V} + 5 \text{ V} = 10.89 \text{ V}$$

D4.6. If we take the zero reference for potential at infinity, find the potential at (0, 0, 2) caused by this charge configuration in free space (a) 12 nC/m on the line $\rho = 2.5 \text{ m}$, $z = 0$; (b) point charge of 18 nC at (1, 2, -1); (c) 12 nC/m on the line $y = 2.5$, $z = 0$, $-1.0 < x < 1.0$.

$$(a) V(\mathbf{r}) = \int \frac{\rho_L(\mathbf{r}') dL'}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} \quad \text{where } dL' = \rho d\phi$$

Since the distance is converted in rectangular coordinates, we can mentally solve it where $\phi = 0^\circ$.

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(0 - 2.5)^2 + (2 - 0)^2} = 3.20$$

$$V(\mathbf{r}) = \int_0^{2\pi} \frac{12 \times 10^{-9} \rho d\phi}{4\pi\epsilon_0 (3.20)} = \frac{12 \times 10^{-9} \rho}{4\pi\epsilon_0 (3.20)} \int_0^{2\pi} d\phi = \frac{12 \times 10^{-9} \rho}{4\pi\epsilon_0 (3.20)} (\phi)_0^{2\pi} = 529 \text{ V}$$

$$(b) V(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|}$$

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(0 - 1)^2 + (0 - 2)^2 + (2 - (-1))^2} = \sqrt{14}$$

$$V(\mathbf{r}) = \frac{18 \times 10^{-9}}{4\pi\epsilon_0 \sqrt{14}} = 43.2 \text{ V}$$

(c) Same formula in (a) but now $dL' = dx$. Solving the distance $|\mathbf{r} - \mathbf{r}'|$ where x is symmetric,

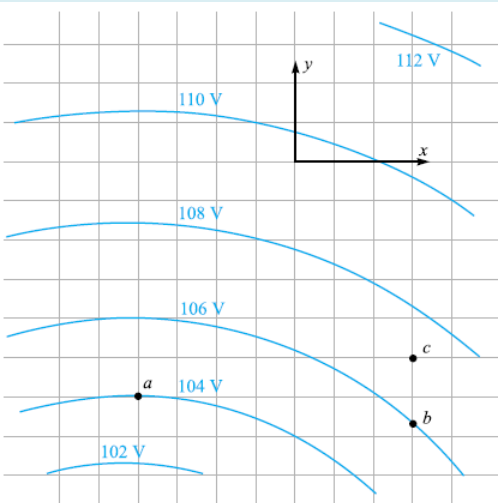
$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(0 - x)^2 + (0 - 2.5)^2 + (2 - 0)^2} = \sqrt{x^2 + 10.25}$$

$$V(\mathbf{r}) = \int_{-1}^1 \frac{12 \times 10^{-9}}{4\pi\epsilon_0 \sqrt{x^2 + 10.25}} dx = \frac{12 \times 10^{-9}}{4\pi\epsilon_0} \int_{-1}^1 \frac{1}{\sqrt{x^2 + 10.25}} dx$$

Integrating $1/\sqrt{x^2 + 10.25}$ is solved by trigonometric substitution and yielding,

$$= \frac{12 \times 10^{-9}}{4\pi\epsilon_0} \left(\ln \left| \frac{x + \sqrt{x^2 + 10.25}}{\sqrt{10.25}} \right| \right)_{-1}^1 = \frac{12 \times 10^{-9}}{4\pi\epsilon_0} (0.615) = 66.3 \text{ V}$$

D4.7. A portion of a two-dimensional ($E_z = 0$) potential field is shown the figure below. The grid lines are 1mm apart in the actual field. Determine approximate values for $\mathbf{E}$ in rectangular coordinates at: (a) a ; (b) b ; (c) c .



Requires graphical solution.

D4.8. Given the potential field in cylindrical coordinates, $V = [100/(z^2 + 1)]\rho \cos \phi \text{ V}$, and point P at $\rho = 3 \text{ m}$, $\phi = 60^\circ$, $z = 2 \text{ m}$, find values at P for (a) V ; (b) $\mathbf{E}$; (c) E ; (d) dV/dN ; (e) $\mathbf{a}_N$; (f) ρ_v in free space.

$$(a) V = [100/(z^2 + 1)]\rho \cos \phi = [100/(2^2 + 1)]3 \cos 60^\circ = 30 \text{ V}$$

$$\begin{aligned}
 (b) \mathbf{E} &= -\nabla V = -\left(\frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial \rho} \mathbf{a}_z \right) \\
 &= -\left(\frac{100}{z^2+1} \cos \phi \mathbf{a}_\rho - \frac{100}{z^2+1} \sin \phi \mathbf{a}_\phi - \frac{200z}{(z^2+1)^2} \rho \cos \phi \mathbf{a}_z \right) = -(10\mathbf{a}_\rho - 17.32\mathbf{a}_\phi - 24\mathbf{a}_z) \\
 &= -10\mathbf{a}_\rho + 17.32\mathbf{a}_\phi + 24\mathbf{a}_z \text{ V/m}
 \end{aligned}$$

$$(c) E = \sqrt{(-10)^2 + 17.32^2 + 24^2} = 31.2 \text{ V/m}$$

$$(d) dV/dN = E = 31.2 \text{ V/m}$$

$$(e) \mathbf{E} = -(dV/dN)\mathbf{a}_N$$

$$\mathbf{a}_N = -\frac{\mathbf{E}}{dV/dN} = -\left(\frac{-10\mathbf{a}_\rho + 17.32\mathbf{a}_\phi + 24\mathbf{a}_z}{31.2} \right) = 0.32\mathbf{a}_\rho - 0.55\mathbf{a}_\phi - 0.77\mathbf{a}_z$$

$$\begin{aligned}
 (f) \mathbf{D} &= \epsilon_0 \mathbf{E} = \epsilon_0 \left(-\frac{100}{z^2+1} \cos \phi \mathbf{a}_\rho + \frac{100}{z^2+1} \sin \phi \mathbf{a}_\phi + \frac{200z}{(z^2+1)^2} \rho \cos \phi \mathbf{a}_z \right) \\
 \rho_v &= \nabla \cdot \mathbf{D} = \text{div } \mathbf{D} = (1/\rho) \partial(\rho D_\rho)/\partial \rho + (1/\rho) \partial D_\phi/\partial \phi + \partial D_z/\partial z \\
 &= 0 + 0 + 200\epsilon_0 \rho \cos \phi \frac{\partial [z/(z^2+1)^2]}{\partial z}
 \end{aligned}$$

Applying product rule of the derivative of $z/(z^2+1)^2$ and yielding,

$$= 200\epsilon_0 \rho \cos \phi \left[\frac{1}{(z^2+1)^2} - \frac{4z^2}{(z^2+1)^3} \right]_{\rho=3, \phi=60^\circ, z=2} = -234 \text{ pC/m}^3$$

D4.9. An electric dipole located at the origin in free space has a moment $\mathbf{p} = 3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z \text{ nC} \cdot \text{m}$. (a) Find V at $P_A(2, 3, 4)$. (b) Find V at $r = 2.5$, $\theta = 30^\circ$, $\phi = 40^\circ$.

$$\begin{aligned}
 (a) V &= \frac{1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \mathbf{p} \cdot \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3} \mathbf{p} \cdot \mathbf{r} - \mathbf{r}' \\
 \mathbf{r} - \mathbf{r}' &= (2-0)\mathbf{a}_x + (3-0)\mathbf{a}_y + (4-0)\mathbf{a}_z = 2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z \\
 |\mathbf{r} - \mathbf{r}'| &= \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29} \\
 V &= \frac{1}{4\pi\epsilon_0 (\sqrt{29})^3} (3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z)(1 \times 10^{-9}) \cdot (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) = \frac{4 \times 10^{-9}}{4\pi\epsilon_0 (\sqrt{29})^3} = 0.23 \text{ V}
 \end{aligned}$$

(b) Using Table 5 from Chapter 1 to convert the point P_A to rectangular coordinates,

$$\begin{aligned}
 x &= 0.958, y = 0.803, z = 2.165 \\
 \mathbf{r} - \mathbf{r}' &= (0.958-0)\mathbf{a}_x + (0.803-0)\mathbf{a}_y + (2.165-0)\mathbf{a}_z = 0.958\mathbf{a}_x + 0.803\mathbf{a}_y + 2.165\mathbf{a}_z \\
 |\mathbf{r} - \mathbf{r}'| &= \sqrt{0.958^2 + 0.803^2 + 2.165^2} = 2.5 \\
 V &= \frac{(3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z)(1 \times 10^{-9})}{4\pi\epsilon_0 2.5^3} \cdot (0.958\mathbf{a}_x + 0.803\mathbf{a}_y + 2.165\mathbf{a}_z) = \frac{3.43 \times 10^{-9}}{4\pi\epsilon_0 2.5^3} = 1.97 \text{ V}
 \end{aligned}$$

D4.10. A dipole of moment $\mathbf{p} = 6\mathbf{a}_z \text{ nC} \cdot \text{m}$ is located at the origin in free space. (a) Find V at $P(r = 4, \theta = 20^\circ, \phi = 0^\circ)$. (b) Find $\mathbf{E}$ at P .

(a) Same formula in D4.9(a). Converting the point P to rectangular coordinates,

$$\begin{aligned}
 x &= 1.368, y = 0, z = 3.759 \\
 \text{Since the procedure is very similar with D4.9(a) and (b) where } \mathbf{p} \text{ is at the origin,} \\
 \mathbf{r} - \mathbf{r}' &= 1.386\mathbf{a}_x + 3.759\mathbf{a}_z \text{ and } |\mathbf{r} - \mathbf{r}'| = 4 \\
 V &= \frac{(1.368\mathbf{a}_x + 3.759\mathbf{a}_z)(1 \times 10^{-9})}{4\pi\epsilon_0 4^3} \cdot 6\mathbf{a}_z = \frac{22.55 \times 10^{-9}}{4^4 \pi\epsilon_0} = 3.167 \text{ V} = 3.17 \text{ V}
 \end{aligned}$$

$$(b) \mathbf{E} = \frac{Qd}{4\pi\epsilon_0 r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)$$

Since there is no charge given, we have to derive with the help of the voltage V since we have the answer in (a). Deriving,

$$V = \frac{Qd \cos \theta}{4\pi\epsilon_0 r^2}; Qd = \frac{4\pi\epsilon_0 r^2 V}{\cos \theta}$$

Substituting,

$$\mathbf{E} = \frac{(4\pi\epsilon_0 r^2)/\cos\theta}{4\pi\epsilon_0 r^3} (2\cos\theta \mathbf{a}_r + \sin\theta \mathbf{a}_\theta) = \frac{V}{r} (2\mathbf{a}_r + \tan\theta \mathbf{a}_\theta)$$

All variables are given in (a). Solving,

$$= \frac{3.167}{4} (2\mathbf{a}_r + \tan 20^\circ \mathbf{a}_\theta) = 1.58\mathbf{a}_r + 0.29\mathbf{a}_\theta \text{ V/m}$$

D4.11. Find the energy stored in free space for the region $2 \text{ mm} < r < 3 \text{ mm}$, $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$, given the potential field $V =$: (a) $200/r \text{ V}$; (b) $(300 \cos \theta)/r^2 \text{ V}$.

$$(a) \mathbf{E} = -\nabla V = -\{(\partial V/\partial r)\mathbf{a}_r + (1/r)(\partial V/\partial \theta)\mathbf{a}_\theta + (1/r \sin \theta)(\partial V/\partial \phi)\mathbf{a}_\phi\}$$

$$= -[(\partial(200/r)/\partial r)\mathbf{a}_r + 0\mathbf{a}_\theta + 0\mathbf{a}_\phi] = (200/r^2)\mathbf{a}_r$$

$$E = \sqrt{(200/r^2)^2} = 200/r^2$$

$$0.5 \int_{\text{vol}} \epsilon_0 E^2 dv = 0.5 \int_{\text{vol}} \epsilon_0 (200/r^2)^2 (r^2 \sin \theta d\theta d\phi dr)$$

$$= 20\,000\epsilon_0 \int_{0.002}^{0.003} (1/r^2) dr \int_0^{\pi/2} \sin \theta d\theta \int_0^{\pi/2} d\phi$$

$$= 20\,000\epsilon_0 \left(-1/r\right)_{0.002}^{0.003} \left(-\cos \theta\right)_0^{\pi/2} \left(\phi\right)_0^{\pi/2} = 20\,000\epsilon_0 (500/3)(\pi/2) = 46.4 \mu\text{J}$$

(b) Same procedure in (a),

$$\mathbf{E} = -\{\partial[(300 \cos \theta)/r^2]/\partial r \mathbf{a}_r + (1/r)\partial[(300 \cos \theta)/r^2]/\partial \theta \mathbf{a}_\theta + 0\mathbf{a}_\phi\}$$

$$= (600 \cos \theta)/r^3 \mathbf{a}_r + (300 \sin \theta)/r^3 \mathbf{a}_\theta$$

$$E = \sqrt{[(600 \cos \theta)/r^3]^2 + [(300 \sin \theta)/r^3]^2} = \sqrt{(360\,000 \cos^2 \theta)/r^6 + (90\,000 \sin^2 \theta)/r^6}$$

$$0.5 \int_{\text{vol}} \epsilon_0 E^2 dv = 0.5 \int_{\text{vol}} \epsilon_0 \{(360\,000 \cos^2 \theta)/r^6 + (90\,000 \sin^2 \theta)/r^6\} (r^2 \sin \theta d\theta d\phi dr)$$

$$= 45\,000\epsilon_0 \int_{\text{vol}} (1/r^4) (4 \sin \theta \cos^2 \theta + \sin^3 \theta) d\theta d\phi dr$$

$$= 45\,000\epsilon_0 \int_{0.002}^{0.003} (1/r^4) dr \int_0^{\pi/2} (4 \sin \theta \cos^2 \theta + \sin^3 \theta) d\theta \int_0^{\pi/2} d\phi$$

Integrating $(4 \sin \theta \cos^2 \theta + \sin^3 \theta)$ is solved by 'integration by substitution' and trigonometric identity yielding,

$$= 45\,000\epsilon_0 \left(-\frac{1}{3r^3}\right)_{0.002}^{0.003} \left(-\cos^3 \theta - \cos \theta\right)_0^{\pi/2} \left(\phi\right)_0^{\pi/2}$$

$$= 45\,000\epsilon_0 [-1/(3(0.003)^3) + 1/(3(0.002)^3)] (2)(\pi/2) = 36.7 \text{ J}$$

CHAPTER 5

D5.1. Given the vector current density $\mathbf{J} = 10\rho^2 z \mathbf{a}_\rho - 4\rho \cos^2 \phi \mathbf{a}_\phi \text{ mA/m}^2$: (a) find the current density at $P(\rho = 3, \phi = 30^\circ, z = 2)$; (b) determine the total current flowing outward through the circular band $\rho = 3$, $0 < \phi < 2\pi$, $2 < z < 2.8$.

$$(a) \mathbf{J} = 10(3^2)(2) \mathbf{a}_\rho - 4(3)(\cos 30^\circ)^2 \mathbf{a}_\phi \text{ mA/m}^2 = 180\mathbf{a}_\rho - 9\mathbf{a}_\phi \text{ mA/m}^2$$

$$(b) I = \oint_S \mathbf{J} \cdot d\mathbf{S} \text{ where } d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$$

$$= (1 \times 10^{-3}) \oint_S (10\rho^2 z \mathbf{a}_\rho - 4\rho \cos^2 \phi \mathbf{a}_\phi) \cdot (\rho d\phi dz \mathbf{a}_\rho + 0\mathbf{a}_\phi) = (1 \times 10^{-3}) 10\rho^3 \int_0^{2\pi} d\phi \int_2^{2.8} z dz$$

$$= 0.01\rho^3 \left(\phi\right)_0^{2\pi} \left(z^2/2\right)_2^{2.8} = 0.01(27)(2\pi)(1.92) = 3.26 \text{ A}$$

D5.2. Current density is given in cylindrical coordinates as $\mathbf{J} = -10^6 z^{1.5} \mathbf{a}_z \text{ A/m}^2$ in the region $0 \leq \rho \leq 20 \mu\text{m}$, $\mathbf{J} = 0$. (a) Find the total current crossing the surface $z = 0.1 \text{ m}$ in the $\mathbf{a}_z$ direction. (b) If the charge velocity is $2 \times 10^6 \text{ m/s}$ at $z = 0.1 \text{ m}$, find ρ_v there. (c) If the volume charge density at $z = 0.15 \text{ m}$ is -2000 C/m^3 , find the charge velocity.

$$(a) I = \oint_S \mathbf{J} \cdot d\mathbf{S} \text{ where } d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$$

$$= \oint_S -10^6 z^{1.5} \mathbf{a}_z \cdot \rho d\phi d\rho \mathbf{a}_z = -10^6 z^{1.5} \int_0^{20 \times 10^{-6}} \rho d\rho \int_0^{2\pi} d\phi$$

$$= -10^6 (0.1^{1.5}) (2 \times 10^{-10}) (2\pi) = -39.7 \mu\text{A}$$

(b) $\mathbf{J} = \rho_v \mathbf{v}$; $\rho_v = J_z/v_z$ where they represent as z component.

$$\rho_v = [-10^6(0.1^{1.5})]/(2 \times 10^6) = -15.8 \text{ mC/m}^3$$

(c) Same formula in (b),

$$v_z = J_z/\rho_v = [-10^6(0.15^{1.5})]/(-2000) = 29.0 \text{ m/s}$$

D5.3. Find the magnitude of the current density in a sample of silver for which $\sigma = 6.17 \times 10^7 \text{ S/m}$ and $\mu_e = 0.0056 \text{ m}^2/\text{V} \cdot \text{s}$ if (a) the drift velocity is $1.5 \text{ } \mu\text{m/s}$; (b) the electric field intensity is 1 mV/m ; (c) the sample is a cube 2.5 mm on a side having a voltage of 0.4 mV between opposite faces; (d) the sample is a cube 2.5 mm on a side carrying a total current of 0.5 A .

(a) $\mathbf{J} = \sigma \mathbf{E}$

Since $\mathbf{v}_d = -\mu_e \mathbf{E}$, then $\mathbf{J} = -\sigma \mathbf{v}_d / \mu_e$.

Since we only find the value of the magnitude of the current density J , which we know that vectors are denoted by bold letters, we consider $\mathbf{J}$ and other vectors as the same dot unit vector. The negative is canceled because they refer to the direction.

$$J = \sigma v_d / \mu_e = [(6.17 \times 10^7)(1.5 \times 10^{-6})]/0.0056 = 16.5 \text{ kA/m}^2$$

$$(b) J = \sigma E = (6.17 \times 10^7)(1 \times 10^{-3}) = 61.7 \text{ kA/m}^2$$

$$(c) J = \sigma V / L = [(6.17 \times 10^7)(0.4 \times 10^{-3})]/(2.5 \times 10^{-3}) = 9.9 \text{ MA/m}^2$$

$$(d) J = I / S = 0.5 / (2.5 \times 10^{-3})^2 = 80 \text{ kA/m}^2$$

D5.4. A copper conductor has a diameter of 0.6 in. and it is 1200 ft long. Assume that it carries a total dc current of 50 A . (a) Find the total resistance of the conductor. (b) What current density exists in it? (c) What is the dc voltage between the conductor ends? (d) How much power is dissipated in the wire?

(a) We must convert the English units to metric systems. A 1200 feet is about 365.76 m . A 0.6 in is about 0.01524 m . The radius is half of the diameter; so $r = 7.62 \times 10^{-3} \text{ m}$.

The conductivity of the copper is $5.8 \times 10^7 \text{ S/m}$.

The area of the circle is πr^2 .

$$R = L / \sigma S = 365.76 / [(5.8 \times 10^7) \pi (7.62 \times 10^{-3})^2] = 0.035 \text{ } \Omega$$

$$(b) J = I / S = 50 / [\pi (7.62 \times 10^{-3})^2] = 2.74 \times 10^5 \text{ A/m}^2$$

$$(c) V = IR = 50(0.035) = 1.75 \text{ V}$$

With regards from the answer of the textbook, $V = 50(0.03457) = 1.73 \text{ V}$

$$(d) P = VI = 1.75(50) = 87.5 \text{ W}$$

With regards from the answer of the textbook, $P = 1.7285(50) = 86.4 \text{ W}$

D5.5. Given the potential field in free space, $V = 100 \sinh 5x \sin 5y \text{ V}$, and a point $P(0.1, 0.2, 0.3)$, find at P : (a) V ; (b) $\mathbf{E}$; (c) $|\mathbf{E}|$; (d) $|\rho_s|$ if it is known that P lies on a conductor surface.

(a) The function $\sinh(x)$ is kind of a hyperbolic trigonometry where $\sinh(x) = (e^x - e^{-x})/2$. Be careful that the points are in radians, not degrees.

$$V = 100 \sinh 5(0.1) \sin 5(0.2) = 43.8 \text{ V}$$

(b) Using back the formula in Chapter 4,

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\{(\partial V / \partial x) \mathbf{a}_x + (\partial V / \partial y) \mathbf{a}_y + (\partial V / \partial z) \mathbf{a}_z\} \\ &= -\{(\partial(100 \sinh 5x \sin 5y) / \partial x) \mathbf{a}_x + (\partial(100 \sinh 5x \sin 5y) / \partial y) \mathbf{a}_y + 0 \mathbf{a}_z\} \end{aligned}$$

The derivative of $\sinh x$ is $\cosh x$ or $(e^x + e^{-x})/2$. But $\sinh 5x \, dx$ is $5 \cosh 5x$.

$$\begin{aligned} &= -500 \cosh 5x \sin 5y \mathbf{a}_x - 500 \sinh 5x \cos 5y \mathbf{a}_y \\ &= -500 \cosh 5(0.1) \sin 5(0.2) \mathbf{a}_x - 500 \sinh 5(0.1) \cos 5(0.2) \mathbf{a}_y \\ &= -474 \mathbf{a}_x - 140.8 \mathbf{a}_y \text{ V/m} \end{aligned}$$

$$(c) E = \sqrt{(-474)^2 + (-140.8)^2} = 495 \text{ V/m}$$

$$(d) \rho_s = \epsilon_0 E_N = \epsilon_0(495) = 4.38 \text{ nC/m}^2$$

D5.6. A perfectly conducting plane is located in free space at $x = 4$, and a uniform infinite line charge of 40 nC/m lies along the line $x = 6, y = 3$. Let $V = 0$ at the conducting plane. At $P(7, -1, 5)$ find: (a) V ; (b) $\mathbf{E}$.

It is really good if we refer to answer first in (b) because it is difficult to solve a voltage with lack of given variables like the charge Q and the electric field $\mathbf{E}$. In (b), a method of image is required. We are going to mirror the infinite line charge ρ_L including the distance (or radial vector) $\mathbf{R}$ without

the conducting plane. For more details, see Section 5.5. The book is explained that the radial vector from positive (or negative if ever) line charge to point P is symmetric where the plane is located. For example, if the line charge (not the plane) is at $z = 3$ and the plane is at $z = 0$, then we put an image line charge at $z = -3$ which the plane is removed.

(b) Locating the radial vector $\mathbf{R}$ from the given location of line charge and P ,

$$\mathbf{R}_+ = \mathbf{a}_x - 4\mathbf{a}_y \text{ and}$$

$$\mathbf{R}_- = -5\mathbf{a}_x - 4\mathbf{a}_y$$

How did we obtain them? Well, draw a two-dimensional rectangular coordinates. Locate P at $x = 7$ and $y = -1$. Notice that no z axis is included because the conducting plane activates on x axis and the line charge activates on x and y axes. The plane is at $x = 4$ while the line charge is $x = 6$ and $y = 3$. Therefore, the *image* line charge is symmetrically at $x = 2$ and $y = 3$. We draw the plane, line charge and image line charge with horizontal line.

We can now find the radial vector $\mathbf{R}$. For $\mathbf{R}_+$, the line charge is one unit away from $P(x = 7)$, so $7 - 6 = 1$; the line charge is four units away from $P(y = 3)$, so $-1 - 3 = -4$. The image line charge reacts on negative distance because we deal with directions no matter where plane is placed. For $\mathbf{R}_-$, the image line charge is five units away from $P(x = 7)$, so $-(7 - 2) = -5$; the image line charge is four units away from $P(y = 3)$ is $-(-1 - 3) = 4$. Finding their magnitudes, $R_+ = \sqrt{17} = 4.123$ and $R_- = \sqrt{41} = 6.403$.

Knowing each electric fields,

$$\mathbf{E}_+ = \frac{\rho_L}{2\pi\epsilon_0 R_+} \mathbf{a}_{R_+} = \frac{40 \times 10^{-9}}{2\pi\epsilon_0 (\sqrt{17})^2} (\mathbf{a}_x - 4\mathbf{a}_y) = 42.294\mathbf{a}_x - 169.177\mathbf{a}_y$$

$$\mathbf{E}_- = \frac{\rho_L}{2\pi\epsilon_0 R_-} \mathbf{a}_{R_-} = \frac{40 \times 10^{-9}}{2\pi\epsilon_0 (\sqrt{41})^2} (-5\mathbf{a}_x + 4\mathbf{a}_y) = -87.683\mathbf{a}_x + 70.147\mathbf{a}_y$$

We cannot get the exact answer from the textbook. Adding together,

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = -45.39\mathbf{a}_x - 99.03\mathbf{a}_y \text{ V/m}$$

(a) We know that the plane stands at x axis. Finding the voltage V where only the electric field E and length for P are extracted at x direction,

$$V = E_x L_x = \sqrt{(-45.39)^2} (7) = 317.73 \text{ V}$$

D5.7. Using the values given in this section for the electron and hole mobilities in silicon at 300 K, and assuming hole and electron charge densities are 0.0029 C/m^3 and -0.0029 C/m^3 , respectively, find: (a) the component of the conductivity due to holes; (b) the component of the conductivity due to electrons; (c) the conductivity.

The electron and hole mobilities of pure silicon is 0.12 and 0.025.

$$(a) \sigma_h = \rho_h \mu_h = (0.0029)(0.025) = 72.5 \text{ } \mu\text{S/m}$$

$$(b) \sigma_e = -\rho_e \mu_e = -(-0.0029)(0.12) = 348 \text{ } \mu\text{S/m}$$

$$(c) \sigma = \sigma_h + \sigma_e = 421 \text{ } \mu\text{S/m}$$

D5.8. A slab of dielectric material has a relative dielectric constant of 3.8 and contains a uniform electric flux density of 8 nC/m^2 . If the material is lossless, find: (a) E ; (b) P ; (c) the average number of dipoles per cubic meter if the average dipole moment is $10^{-29} \text{ C} \cdot \text{m}$.

$$(a) \mathbf{D} = \epsilon \mathbf{E} \text{ where } \epsilon = \epsilon_0 \epsilon_r$$

$$E = D/\epsilon_0 \epsilon_r = (8 \times 10^{-9})/3.8\epsilon_0 = 238 \text{ V/m}$$

$$(b) \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}; P = D - \epsilon_0 E$$

$$P = (8 \times 10^{-9}) - 238\epsilon_0 = 5.89 \text{ nC/m}^2$$

$$(c) \mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^{n\Delta v} \mathbf{p}_i$$

We have one dipole moment $\mathbf{p}$ and no limit of the average volume Δv . Generalizing the formula,

$$\mathbf{P} = \mathbf{p}/\Delta v; 1/\Delta v = P/p = 5.89 \times 10^{-9}/10^{-29} = 5.89 \times 10^{20} \text{ m}^{-3}$$

D5.9. Let Region 1 ($z < 0$) be composed of a uniform dielectric material for which $\epsilon_r = 3.2$, while Region 2 ($z > 0$) is characterized by $\epsilon_r = 2$. Let $\mathbf{D}_1 = -30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z \text{ nC/m}^2$ and find: (a) D_M ; (b) $\mathbf{D}_{t1}$; (c) D_{t1} ; (d) D_1 ; (e) θ_1 ; (f) $\mathbf{P}_1$.

(a) Since the Region 1 affects the z axis, only D_z is referred.

$$D_{N1} = D_{1z} = \sqrt{70^2} \text{ nC/m}^2 = 70 \text{ nC/m}^2$$

(b) While the normal component D_N refers to the mentioned region where it locates, then the rest which does not affect the region is D_{t1} .

$$\mathbf{D}_{t1} = \mathbf{D}_1 - \mathbf{D}_{N1} = [(-30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z) - 70\mathbf{a}_z] \text{ nC/m}^2 = -30\mathbf{a}_x + 50\mathbf{a}_y \text{ nC/m}^2$$

$$(c) D_{t1} = \sqrt{(-30)^2 + 50^2} = 58.31 \text{ nC/m}^2$$

$$(d) D_1 = \sqrt{(-30)^2 + 50^2 + 70^2} = 91.1 \text{ nC/m}^2$$

$$(e) D_{N1} = D_1 \cos \theta_1$$

$$\theta_1 = \cos^{-1} D_{N1}/D_1 = \cos^{-1} (70 \times 10^{-9} / 91.1 \times 10^{-9}) = 39.8^\circ$$

$$(f) \mathbf{D}_1 = \epsilon_0 \mathbf{E}_1 + \mathbf{P}_1 \text{ where } \mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}$$

$$\begin{aligned} \mathbf{P}_1 &= \mathbf{D}_1 - \epsilon_0 \mathbf{E}_1 = \mathbf{D}_1(1 - (1/\epsilon_r)) = (-30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z)(1 - (1/3.2)) \text{ nC/m}^2 \\ &= -20.6\mathbf{a}_x + 34.4\mathbf{a}_y + 48.1\mathbf{a}_z \text{ nC/m}^2 \end{aligned}$$

D5.10. Continue Problem D5.9 by finding: (a) $\mathbf{D}_{N2}$; (b) $\mathbf{D}_{t2}$; (c) $\mathbf{D}_2$; (d) $\mathbf{P}_2$; (e) θ_2 .

$$(a) \mathbf{D}_{N1} = \mathbf{D}_{N2} = 70\mathbf{a}_z \text{ nC/m}^2$$

(b) $\mathbf{D}_{t1}$ is not equal to $\mathbf{D}_{t2}$ because of their relative permittivity ϵ_r .

$$\mathbf{D}_{t1}/\mathbf{D}_{t2} = \epsilon_{r1}/\epsilon_{r2}$$

$$\mathbf{D}_{t2} = \mathbf{D}_{t1}\epsilon_{r2}/\epsilon_{r1} = [(-30\mathbf{a}_x + 50\mathbf{a}_y)2] / 3.2 \text{ nC/m}^2 = -18.75\mathbf{a}_x + 31.25\mathbf{a}_y \text{ nC/m}^2$$

(c) $\mathbf{D}_1$ is not equal to $\mathbf{D}_2$ because $\mathbf{D}_2$ is the cause of the refraction.

$$\mathbf{D}_2 = \mathbf{D}_{t2} + \mathbf{D}_{N2} = -18.75\mathbf{a}_x + 31.25\mathbf{a}_y + 70\mathbf{a}_z \text{ nC/m}^2$$

$$\begin{aligned} (d) \mathbf{P}_2 &= \mathbf{D}_2(1 - (1/\epsilon_{r2})) = (-18.75\mathbf{a}_x + 31.25\mathbf{a}_y + 70\mathbf{a}_z)(1 - (1/2)) \text{ nC/m}^2 \\ &= -9.38\mathbf{a}_x + 15.63\mathbf{a}_y + 35\mathbf{a}_z \text{ nC/m}^2 \end{aligned}$$

$$(e) \tan \theta_1 / \tan \theta_2 = \epsilon_{r1} / \epsilon_{r2}$$

$$\theta_2 = \tan^{-1} [(\epsilon_{r2} \tan \theta_1) / \epsilon_{r1}] = \tan^{-1} [(2 \tan 39.8^\circ) / 3.2] = 27.5^\circ$$

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